

Valence Asymmetries in Attitude Ambivalence

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Existing models of ambivalence suggest that as the number of conflicting reactions (e.g., attitude components) increases, so too does the experience of ambivalence. Interestingly, though, these models overwhelmingly assume that this relationship is independent of valence. Across 3 studies we observe that this effect is in fact heavily influenced by 2 established valence asymmetries: *positivity offset* (baseline positive reactions even in the absence of positive information) and *negativity bias* (greater impact of negative reactions than positive reactions). Consistent with positivity offset, we observe that subjective ambivalence is greater when people have univalent negative rather than univalent positive attitudes. However, as conflicting information is acquired, subjective ambivalence rises more quickly when that information is negative rather than positive. The latter effect is consistent with negativity bias and suggests that although people feel more conflicted when they have only negative (vs. only positive) reactions, they also feel more conflicted when they have *mostly* positive (vs. mostly negative) reactions. Our investigation also uncovers an interesting consequence of these asymmetries: When people have mixed reactions, they do not experience maximum ambivalence at equal levels of positivity and negativity, as suggested by canonical ambivalence theory. Rather, subjective ambivalence peaks when positive reactions outnumber negative reactions. These effects are found to have downstream consequences for other dimensions of attitude strength.

Keywords: attitude strength, attitudes, negativity bias, positivity offset, valence asymmetry

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People often have positive and negative reactions to the same attitude object. Chocolate cake tastes great, but is high in calories; a smartphone might have a large screen, but also be heavy; a friend can be likable, but unreliable. Subjective attitude ambivalence results naturally from the conflict described in these kinds of examples. And this subjective feeling of ambivalence can be profoundly important. Indeed, an extensive literature delineates the numerous consequences of feeling conflicted based on the simultaneous presence of positive and negative reactions to the same attitude object. However, something this literature has failed to fully capture is that the effect of positive and negative reactions can be asymmetrical. In particular, negative reactions are often stronger than positive reactions (*negativity bias*), but there is also a tendency toward positive reactions even in the absence of any positive information (*positivity offset*). Despite widespread knowledge of these valence asymmetries in judgment and evaluation (Baumeister, Bratslavsky, Finkenauer, & Vohs, 2001; Cacioppo, Gardner, & Berntson, 1997; Fazio, Pietri, Rocklage, & Shook, 2015; Norris, Gollan, Berntson, & Cacioppo, 2010), researchers have yet to give systematic attention to the distinct impact of positive versus negative reactions on subjective ambivalence. The current research seeks to fill this gap. We build on findings from

the literature on valence asymmetries in evaluative judgment—in particular, negativity bias and positivity offset—to better understand attitude ambivalence.

Attitude Ambivalence

Attitude ambivalence results from conflicting positive and negative reactions to the same attitude object (Conner & Armitage, 2008). The study of ambivalence has garnered considerable attention over the years, and has yielded a thorough accounting of both the antecedents (Armitage & Conner, 2000; DeMarree, Wheeler, Briñol, & Petty, 2014; Gebauer, Maio, & Pakizah, 2013; Priester, Petty, & Park, 2007; Scott, 1966; Thompson, Zanna, & Griffin, 1995) and consequences (Maio, Bell, & Esses, 1996; Priester & Petty, 2001; Sawicki et al., 2013; van Harreveld, Rutjens, Schneider, Nohlen, & Keskinis, 2014) of feeling ambivalent about a particular attitude object. Generally speaking, ambivalence is viewed as an aspect of attitude strength (Krosnick & Petty, 1995; Thompson et al., 1995)—the more ambivalent one feels, the less durable and impactful (e.g., less persistent and resistant to change, less influential over thought and behavior) one's attitude tends to be (see Bassili, 1996; Conner & Sparks, 2002). Therefore, precisely mapping the circumstances that produce ambivalence is important; it allows us to understand when a person's attitude will impact his or her thoughts, feelings, and actions.

As noted, ambivalence often emanates from the coexistence of positive and negative information about (or reactions to) a single attitude object (Jonas, Brömer, & Diehl, 2000; Kaplan, 1972; Katz & Hass, 1988; Thompson et al., 1995). This coexistence, or coactivation, can produce a kind of felt conflict or indecision that is frequently termed *subjective ambivalence* (Conner & Armitage,

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2008; Priester & Petty, 1996). Subjective ambivalence is the primary focus of the current investigation.¹ To study ambivalence (objective and subjective), researchers often create attitudes composed of equal proportions of positive and negative reactions—for example, they provide participants with an equal amount of positive and negative information about the same object or issue. The logic is that, compared with univalent (i.e., single-sided) attitudes, the most subjectively conflicting attitudes would be those composed of equal parts positive and negative. We use the term *balanced* to describe these attitudes. An attitude that leans either more positive or more negative is by definition less mixed than a balanced one.

In the event that perfect balance does not exist—that is, when an attitude skews positive or negative—the reactions or components that are more numerous are described as *dominant* (*D*) and the reactions or components that are less numerous are described as *conflicting* (*C*; Priester & Petty, 1996). For instance, if an individual's attitude toward another person is made up of 3 positive reactions and 1 negative reaction, then the positive reactions are dominant and the negative reaction is conflicting (i.e., 3D, 1C). The general view in ambivalence research is that as conflict increases—shifting, for instance, from 5 dominant and 2 conflicting reactions (5D, 2C) to 5 dominant and 3 conflicting reactions (5D, 3C)—subjective ambivalence increases as well. That is, the more mixed attitude (5D, 3C), with its greater number of conflicting reactions (3C vs. 2C), produces more subjective ambivalence.

Interestingly, despite a voluminous literature on subjective ambivalence and its origins, there is little mention of the possibility that positivity and negativity may play distinct, asymmetrical roles in eliciting this subjective experience of conflict. That is, researchers tend to emphasize the relative number of dominant versus conflicting reactions as a determinant of subjective ambivalence, with little consideration of whether those dominant and conflicting reactions are positive or negative. This is surprising given the vast body of work on valence asymmetries in attitudes and social judgment more generally. Indeed, it is well-documented that positive and negative information do not carry equal weight in evaluation, yet this insight has had little impact on ambivalence research. We submit that understanding the unique effects of positivity versus negativity on ambivalence—in particular, on the subjective conflict one experiences when one has positive and negative reactions to the same attitude object—could greatly expand our understanding of ambivalence and the instances in which it arises.

Valence Asymmetries

In their now classic work on the Evaluative Space Model (ESM), Cacioppo et al. (1997) synthesized prior research to describe the operation of two complementary forms of bias for negative and positive information that result in attitude valence asymmetries. These include negativity bias and positivity offset. Briefly, *negativity bias* refers to the fact that, all else equal, negative information generally carries more psychological weight or has greater impact on attitudes than does positive information. *Positivity offset* refers to the idea that when people have no, neutral, or nondiagnostic information about something (or otherwise lack evidence for positive reactions), their default attitudes tend to be slightly favorable. These asymmetries have received

little attention in ambivalence research, but they have been studied extensively in other areas.

Indeed, the negativity bias is believed to be a general phenomenon whereby people place greater weight on negative evaluations than on otherwise equivalent positive evaluations (Baumeister et al., 2001; Rozin & Royzman, 2001). Prospect theory, for example, describes a loss aversion that is greater in magnitude than the preference for equal gain (Kahneman & Tversky, 1979). This effect is seen in the domain of attitudes as well. For instance, compared with positive attitudes, negative attitudes are formed more quickly and transfer to a new target more readily (e.g., Fazio, Eiser, & Shook, 2004; Ratliff & Nosek, 2011). Furthermore, negatively framed beliefs and attitudes (e.g., what one opposes rather than supports) have been shown to feel more important and be more resistant to change than their positively framed counterparts (Bizer & Petty, 2005; Bizer, Larsen, & Petty, 2011; Goodwin & Darley, 2012). Although researchers have found individual variation in the extent to which negative attitudes dominate (Pietri, Fazio, & Shook, 2013; Rocklage & Fazio, 2014), evidence from a variety of sources supports a widespread negativity bias.

In contrast to negativity bias, positivity offset is defined as baseline positive reactions at minimal positive input (Cacioppo et al., 1997; Norris et al., 2010). That is, people often have initial positive attitudes toward entities about which they have no positive information, or only neutral or nondiagnostic information. For example, people often perceive individuals favorably even when they lack knowledge about them (Sears, 1983). Various explanations have been proposed for the source of this positivity, such as generally positive expectations or the adaptive nature of exploratory, approach-oriented behavior in nonthreatening environments (Miller, 1959). Of greatest importance, positivity offset is thought to be most observable when there is a relative absence of positive information. That is, rather than giving extra *weight* to positive information, positivity offset refers to the phenomenon of having a positive attitude despite no positive information. Once people have both positive and negative information, the negative is generally expected to outweigh the positive.

Valence Asymmetries in Attitude Ambivalence

Despite widespread acknowledgment of negativity bias and positivity offset in other areas of judgment and evaluation, these asymmetries have received almost no attention in ambivalence research. The sole exception we could identify is Cacioppo and colleagues' ESM.

Although the primary focus of the ESM is on identifying the roles of positivity offset and negativity bias in attitude *valence*, Cacioppo et al. also speculated about the role these asymmetries might play in ambivalence. In essence, the ESM makes two

¹ Considerable attention in the ambivalence literature has been devoted to identifying formulae that can estimate this subjective experience of ambivalence from people's separable positive and negative reactions to an attitude object. These formulae typically calculate what is referred to as objective (or potential) ambivalence. Interestingly, no existing formulae incorporate either of the known valence asymmetries (negativity bias or positivity offset). The purpose of the current work is not to provide a formulation of objective ambivalence, but rather to investigate whether positivity offset and negativity bias affect the subjective experience of ambivalence.

predictions for ambivalence. First, when there are no conflicting reactions, negativity will be associated with greater ambivalence than positivity. That is, people will feel more ambivalent when they have only negative reactions to an attitude object than when they have only positive reactions. This difference is thought to stem from positivity offset: When people have only negative reactions, "there is still a weak level of positive output" (Cacioppo et al., 1997, p. 15) that conflicts with the negative reactions and creates ambivalence. When people have only positive reactions, there is no inverse negative output. The ESM's second prediction for ambivalence is that when there *are* conflicting reactions, ambivalence will be greater when the dominant reactions are negative rather than positive. In other words, people will feel more ambivalent when they have more negative than positive reactions compared with when they have more positive than negative reactions. This difference is thought to stem from negativity bias and the relatively greater psychological impact (or "motivational output," Cacioppo et al., 1997, p. 15) of negativity compared with positivity.

In short, although the ESM discusses positivity offset and negativity bias as two different asymmetries, it postulates that they have directionally similar effects on ambivalence—in general, negative reactions are proposed to foster greater ambivalence than positive reactions. In the words of the ESM, "although the reason for the asymmetry is different, the nature of the asymmetry is similar: Attitude ambivalence should . . . be more closely associated with the activation of negativity than positivity" (Cacioppo et al., 1997, pp. 15–16). This can be viewed as a main effect prediction for the influence of negativity versus positivity on ambivalence in that ambivalence is always predicted to be greater in response to negative than positive reactions. Indeed, the ESM proposes that positivity offset and negativity bias are complementary. One or the other operates, depending on how mixed people's information or reactions are, but they push ambivalence in a similar direction.

In contrast, we predict an interaction based on the notion that these asymmetries can have more opposing effects. As a starting point, we agree that when there are no conflicting reactions, people will feel more ambivalent when they have negative rather than positive reactions. This prediction is identical to that of the ESM and is based on positivity offset. Importantly, though, when there *are* conflicting reactions, our prediction diverges. In this case, we posit that ambivalence will be greater when the dominant reactions are positive rather than negative. Stated differently, when people have conflicting reactions, we submit that they will feel more conflicted when those *conflicting* reactions are *negative*. Again, negativity bias amplifies the effect of negative relative to positive reactions. This means that when negative reactions are numerically dominant, they should become even more psychologically dominant (leaving a person less ambivalent) given their added weight. Conversely, when negative reactions are numerically conflicting, they should create even more psychological conflict (leaving a person more ambivalent) given their added weight. Thus, although we share the first prediction of the ESM as it pertains to ambivalence, we suggest that the second prediction is likely to be incorrect.

Consider an example in which an individual has either 8 positive and 4 negative reactions or 4 positive and 8 negative reactions to the same stimulus. The dominant view in ambivalence research is

that these cases would produce equivalent levels of ambivalence given that in each case the individual has 8 dominant and 4 conflicting reactions. The ESM, by contrast, suggests that the person would feel more ambivalent in the second case—that is, ambivalence would be greater when a person has 4 positive and 8 negative reactions rather than 8 positive and 4 negative reactions. We depart from both views and contend that ambivalence is greater in the case of 8 positive and 4 negative reactions, because the extra weight given to negative information would begin to equate the psychological magnitude of positive and negative reactions.

Imagine a situation in which someone has only positive or only negative reactions to an object, and then receives new information of the opposite valence. For instance, suppose a person has 8 positive and 0 negative pieces of information about something and then starts receiving negative information, one piece at a time. Another person has 8 negative and 0 positive pieces of information and then starts receiving positive information. If negative information carries more weight than positive, then adding negative information to a positive attitude should produce a quicker, or steeper, rise in ambivalence than adding positive information to a negative attitude.

If we were to plot the effect on subjective ambivalence of adding negative or positive conflicting reactions to a mostly positive or negative attitude, respectively, negativity bias dictates that we should observe a steeper slope in response to adding negative conflicting reactions. The result: Although the person might initially be more ambivalent when she has 8 negative and 0 positive reactions than vice versa (as a result of positivity offset), the steeper slope associated with adding negative conflicting reactions (as a result of negativity bias) would eventually reverse this pattern and result in greater ambivalence associated with negative rather than positive conflicting reactions. In other words, as illustrated in Figure 1, positivity offset should produce a higher y-intercept when there are zero positive (vs. zero negative) conflicting reactions, but *adding* conflicting reactions should produce a steeper rise in felt conflict when those conflicting reactions are negative rather than positive.

In the current research, we will investigate these issues by examining both the y-intercepts and the slopes for the effect of adding negative versus positive conflicting reactions on subjective ambivalence (for similar uses of intercepts and slopes to assess positivity offset and negativity bias, see Ito & Cacioppo, 2005; Ito, Cacioppo, & Lang, 1998; Ito, Larsen, Smith, & Cacioppo, 1998). Note that these tests were not possible in the original ESM work (Cacioppo et al., 1997), which focused primarily on comparing positive reactions only to negative reactions only in their effects on ambivalence. Cacioppo et al. did include a balanced condition in which participants received equal proportions of positive and negative information, but there were no other mixed conditions in their design using different proportions of positive and negative reactions, which are necessary to fully assess the above hypotheses.

The Current Research

In short, the current work extends the principles of negativity bias and positivity offset to the study of ambivalence. We formalize our predictions in contrast to a classic, valence-agnostic un-

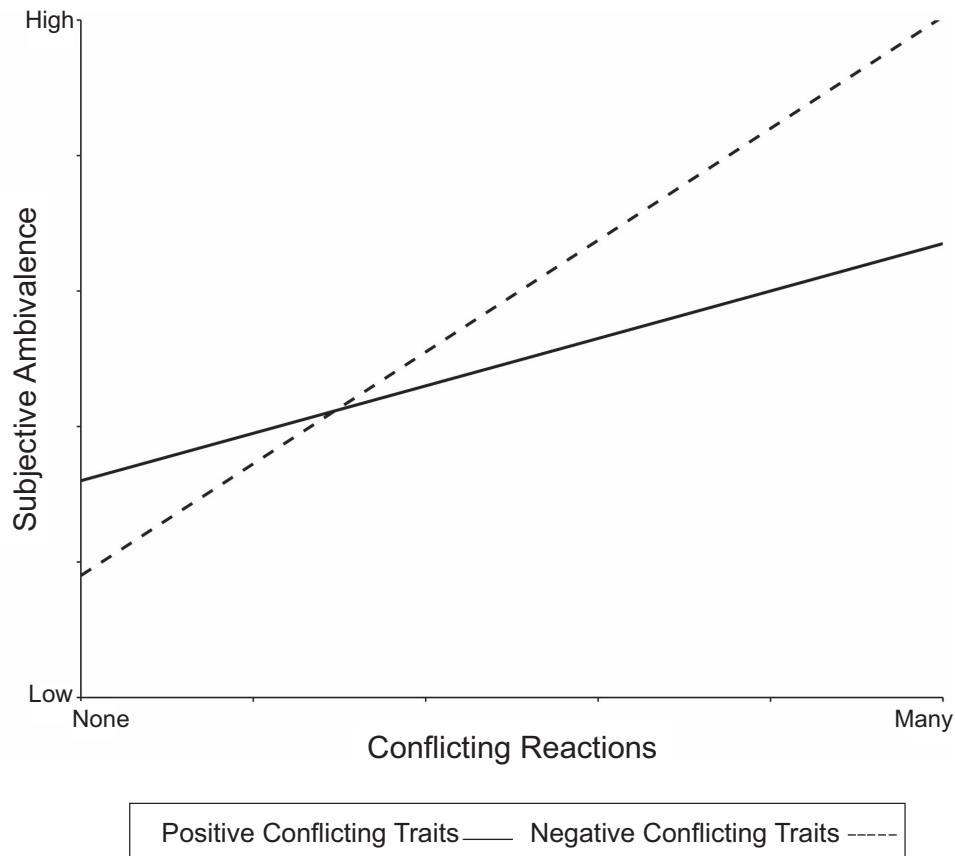


Figure 1. Predictions regarding the effect of adding conflicting positive or negative reactions to a negative or positive dominant attitude, respectively.

derstanding of subjective ambivalence suggesting that ambivalence increases with coactivation of negative and positive reactions. We expect subjective ambivalence to depart from conventional predictions in three major ways. First, when people have univalent negative attitudes, positivity offset should foster some default level of positivity that conflicts with those attitudes and creates ambivalence. When people have univalent positive attitudes, however, positivity offset would only add to that positivity. Thus, ambivalence will be higher when people have only negative rather than only positive reactions. This prediction is consistent with the ESM. Second, as positive and negative reactions become more mixed (i.e., less univalent), we postulate that subjective ambivalence should be greater when conflicting reactions are negative rather than positive. Similarly, adding conflicting negative reactions to a positive attitude should produce a steeper rise in ambivalence than adding conflicting positive reactions to a negative attitude. These findings would be consistent with a negativity bias, but depart from the predictions of the ESM. Third, we hypothesize that maximum ambivalence will be felt not when positive and negative reactions are present in equal proportion, as past research has assumed, but rather when positive reactions slightly outnumber negative reactions.

These predictions are represented visually in Figure 2 as a series of density plots showing subjective ambivalence as a function of positive and negative reactions (darker shading =

greater subjective ambivalence). In each figure panel, negative reactions are plotted on the x axis and positive reactions on the y axis. Thus, the diagonal line represents cases of balanced attitudes; points above the diagonal represent cases of negative conflicting, positive dominant attitudes; and points below the diagonal represent cases of positive conflicting, negative dominant attitudes. The panels themselves reflect predictions made by (A) classic models of ambivalence, (B) a model incorporating only negativity bias, (C) a model incorporating only positivity offset, and (D) a model incorporating *both* negativity bias and positivity offset (the focus of this work).

Starting with classic models of ambivalence (Figure 2A), we observe several notable features: First, subjective ambivalence is lowest in the absence of both positive and negative reactions. Ambivalence is also minimal for only positive reactions (cells adjacent to the y axis) or only negative reactions (cells adjacent to the x axis). Ambivalence begins to manifest as positive and negative reactions present simultaneously (i.e., become coactive). Maximum ambivalence follows the *coactivity diagonal* (depicted by a solid black line), meaning that people feel most ambivalent when they have equivalent positive and negative reactions. Moving perpendicular to any point on the coactivity diagonal, as the distance from the diagonal increases, ambivalence reduces symmetrically (i.e., without valence asymmetry).

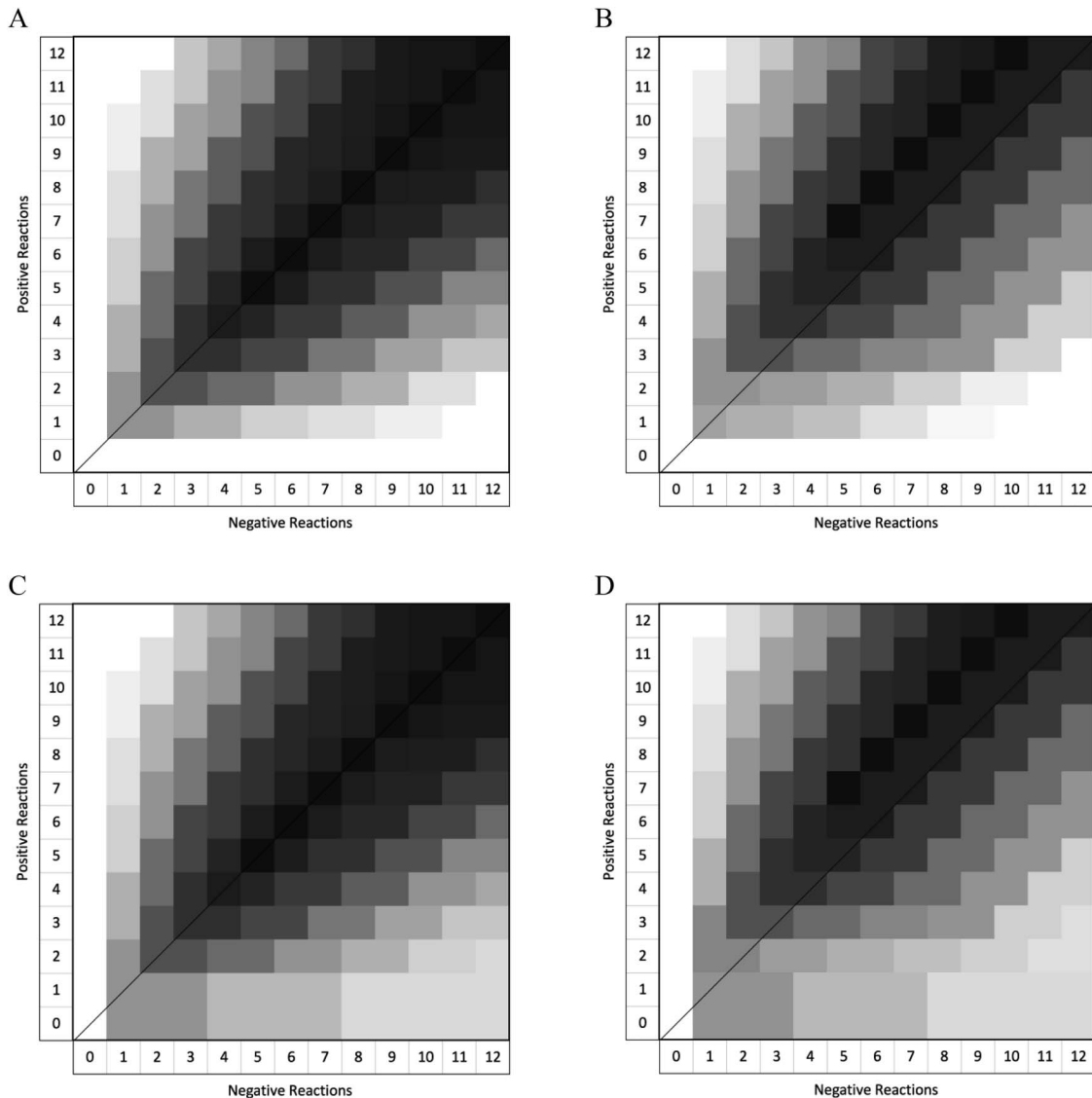


Figure 2. Density plots for subjective ambivalence as a function of negative reactions (x axis) and positive reactions (y axis). Darker shades indicate higher subjective ambivalence. (A) Classic models show ambivalence distributed evenly about the coactivity diagonal. (B) Negativity bias increases conflict on the positive dominant (negative conflicting) side of the diagonal. (C) Positivity offset increases conflict when only negative reactions exist. (D) Simulated representation of the combined effect of negativity bias and positivity offset on subjective ambivalence.

Although the above features are consistent with classic models of ambivalence, none of them reflects negativity bias or positivity offset. Figure 2B and 2C incorporates these individual asymmetries. Figure 2B captures negativity bias. Here, we have increased the relative weight of negative reactions, thus shifting the gradient slightly upward. As a result, maximum ambivalence occurs not when positive and negative reactions are balanced, but rather when positive reactions outnumber negative reactions. Note also that ambivalence is more sensitive to conflicting reactions that are negative (vs. positive). That is, there is a steeper rise in subjective ambivalence as we increase the number of negative rather than positive conflicting

reactions. Figure 2C captures positivity offset. Importantly, positivity offset does not shift the gradient, but rather suggests a valence asymmetry along the axes. In this case, as reactions become more univalent, positivity offset increases the degree of subjective ambivalence people feel when conflicting reactions are positive rather than negative. In other words, highly dominant negative attitudes (e.g., with no conflicting positive reactions) feel more ambivalent than highly dominant positive attitudes (e.g., with no conflicting negative reactions). Figure 2D incorporates both positivity offset and negativity bias into a new, more complete map of subjective ambivalence. It is this map that we test in the current research.

Study 1

Study 1 was designed to provide an initial test of positivity offset and negativity bias in ambivalence across a variety of issues. Specifically, we conducted a large-scale survey with several thousand participants, resulting in 16,167 analyzable attitudes on a diversity of topics. In addition to attitudes, we assessed separate positive and negative reactions (i.e., attitude positivity and negativity) along with subjective ambivalence. For analysis, each attitude was classified according to its conflicting valence (i.e., whether positive or negative reactions were less numerous) and the degree to which the dominant and conflicting reactions were mixed (low, moderate, or high; see below for classification scheme). This permitted us to test for the operation of negativity bias and positivity offset in attitudes that were relatively more mixed (i.e., more balanced) versus less mixed (i.e., more univalent). As noted earlier (and illustrated in Figure 2), we hypothesized that we would observe greater subjective ambivalence when conflicting reactions were negative and dominant reactions were positive rather than when conflicting reactions were positive and dominant reactions were negative. This finding would be consistent with negativity bias. Support for positivity offset would come from an attenuation or even reversal of this asymmetry for univalent attitudes.

Method

Participants and procedure. A total of 4,098 individuals (49.9% Female; $M_{\text{age}} = 34.3$; $SD_{\text{age}} = 11.5$) from multiple online populations participated for direct or indirect compensation.² Each participant completed dependent measures for 2, 5, or 7 attitude issues (depending on sample), resulting in 16,167 unique attitude observations (the unit of analysis). At the study outset, participants read they would provide their opinions on a variety of issues that would be presented as either statements or topics. For each issue, participants responded to a global attitude item, an attitude positivity item (to assess positive reactions), an attitude negativity item (to assess negative reactions), and 3 subjective ambivalence items.

Dependent measures.

Attitudes. Attitude issues were drawn from the literature on ambivalence and included the following topics: reusable water bottles, abolition of K–12 summer break, mandatory health insurance, accessible condoms for students, junk food tax, pilotless commercial airline flights, pay-for-use public roads, tabloid magazines, hamburgers, neutral-toned wall paint, one's mother, raising college tuition, legalized abortion, and affirmative action. Immediately following the presentation of each issue, participants were asked "How would you describe your thoughts and feelings about this topic?" 1 = *Strongly Negative*, 7 = *Strongly Positive*; or for each statement: "Please indicate how much you agree or disagree with this statement." 1 = *Strongly Disagree*, 7 = *Strongly Agree*.

Positive and negative reactions. Next, for each issue, participants answered an attitude positivity item to assess positive reactions ("Considering only your POSITIVE thoughts and feelings about this topic and ignoring the negative ones, how positive would you say your positive thoughts and feelings are?" 0 = *No positive thoughts or feelings*, 9 = *Maximum positive thoughts or feelings*) and an attitude negativity item to assess negative reactions ("Considering only your NEGATIVE thoughts and feelings about this topic and ignoring the positive ones, how negative

would you say your negative thoughts and feelings are?" 0 = *No negative thoughts or feelings*, 9 = *Maximum negative thoughts or feelings*). These measures of positive and negative reactions (adapted from Priester & Petty, 1996) were randomly ordered for each participant.

Subjective ambivalence. Finally, we assessed subjective ambivalence for each issue using the following three items (adapted from Priester & Petty, 1996): "To what extent do you feel undecided about how good or bad [this topic] is?" (1 = *Not At All Undecided*, 7 = *Very Undecided*), "To what extent do you feel conflicted about how good or bad [this topic] is?" (1 = *Not At All Conflicted*, 7 = *Very Conflicted*), and "To what extent do you have mixed feelings about how good or bad [this topic] is?" (1 = *Not At All Mixed*, 7 = *Very Mixed*).³ Responses were averaged to form a composite index for each topic ($\alpha = .94$).

Analysis coding. First, we examined whether the valence of conflicting reactions affected the subjective ambivalence associated with the attitude. To do this, positive and negative reactions were recoded according to whether they were dominant (D) or conflicting (C). For instance, an attitude with 7 positive and 5 negative reactions (or an attitude with 5 positive and 7 negative reactions) would be coded as 7 dominant and 5 conflicting reactions. The valence of conflicting reactions was then recorded. Thus, an attitude with 7 positive and 5 negative reactions and an attitude with 7 negative and 5 positive reactions would both be coded as 7D and 5C, but they would differ in their conflicting reaction valence—negative and positive, respectively.

Next, we categorized attitudes based on their degree of mixture, or objective conflict (high, moderate, or low). Attitudes in which dominant reactions outnumbered conflicting reactions by a magnitude of only 1–3 were coded as high conflict; 4–6 as moderate conflict; and 7–9 as low conflict. For instance, an attitude with 7 positive and 5 negative reactions (7D, 5C; difference = 2, conflicting valence = negative) would be coded as high conflict, whereas an attitude with 2 positive and 9 negative reactions (9D, 2C; difference = 7, conflicting valence = positive) would be coded as low conflict. This coding maps a key tenet of ambivalence theory: More mixed attitudes (i.e., those for which dominant and conflicting reactions are closer in magnitude) are associated with greater subjective ambivalence (Thompson et al., 1995). Because we used participants' separate positive and negative reaction scores as the basis for this categorization, we did not submit them to analysis as a dependent measure (e.g., in the form of objective ambivalence).

² These data were aggregated across two online populations (Amazon's Mechanical Turk and SurveyMonkey's Contribute). Mechanical Turk participants were compensated directly for their time, whereas SurveyMonkey participants were compensated indirectly by selecting a charity to receive an equivalent donation.

³ Although subjective ambivalence for most of these issues was assessed on 7-point scales, on a few issues, drawn from Priester and Petty (1996), we employed 9-point scales. To streamline our presentation and analysis, we transformed subjective ambivalence scores from the 9-point scales to a 7-point scale so that the possible range of responses in our analyses for all issues was 1 to 7. Standardizing and averaging across issues produces equivalent results to those presented here.

Table 1

Study 1: Dependent Measures as a Function of Conflicting Reaction Valence and Conflict Magnitude for Global Attitude, Attitude Positivity, Attitude Negativity, and Subjective Ambivalence

Conflicting valence	Low conflict	Moderate conflict	High conflict	Balanced
Global attitude				
Positive	1.36 (.89) [2,914]	2.27 (1.22) [1,902]	3.37 (1.70) [2,194]	—
Negative	7.33 (1.95) [2,079]	5.72 (1.91) [2,068]	4.81 (1.70) [2,645]	—
Balanced	—	—	—	4.34 (1.84) [2,365]
Attitude positivity				
Positive	.24 (.47) [2,914]	1.81 (.97) [1,902]	3.64 (1.52) [2,194]	—
Negative	8.58 (.61) [2,079]	6.89 (.97) [2,068]	5.68 (1.34) [2,645]	—
Balanced	—	—	—	4.57 (2.32) [2,365]
Attitude negativity				
Positive	8.63 (.59) [2,914]	6.75 (.94) [1,902]	5.60 (1.44) [2,194]	—
Negative	.34 (.54) [2,079]	1.96 (1.01) [2,068]	3.72 (1.38) [2,645]	—
Balanced	—	—	—	4.57 (2.32) [2,365]
Subjective ambivalence				
Positive	1.45 (.97) [2,914]	2.48 (1.30) [1,902]	3.48 (1.47) [2,194]	—
Negative	1.46 (.86) [2,079]	2.71 (1.30) [2,068]	3.65 (1.31) [2,645]	—
Balanced	—	—	—	3.39 (1.60) [2,365]

Note. Standard deviations are included parenthetically, response frequencies are included in brackets. Low conflict attitudes were those where dominant reactions outnumbered conflicting by 7-9, moderate conflict by 4-6, and high conflict by 1-3. Balanced attitudes were characterized by equal numbers of positive and negative reactions.

Results

All cell means, standard deviations, and frequencies are presented in Table 1. A correlation matrix for all dependent measures in Study 1 is included as Table 2. We analyzed the data by conducting a series of mixed-model regression analyses that allowed us to control for the expected variation in ambivalence across attitude issues and participants. The best-fit model included fixed-effects at the level of participant, attitude issue, population, and study (using the lme4 v1.1-7 package in RStudio v0.98.501). As expected, regressing subjective ambivalence on degree of conflict revealed that high conflict attitudes ($M = 3.54$, $SD = 1.60$) were more subjectively ambivalent than moderate conflict attitudes ($M = 2.59$, $SD = 1.30$; $t(15,892) = 35.99$, $p < .001$), which in turn were more subjectively ambivalent than low conflict attitudes ($M = 1.48$, $SD = 0.92$; $t(15,994) = 41.47$, $p < .001$). More germane to our hypothesis, collapsing across all combinations of dominant and conflicting attitude reactions and focusing only on conflicting reaction valence, subjective ambivalence was greater, $t(15,669) = 11.78$, $p < .001$ when conflicting reactions were negative ($M = 2.68$, $SD = 1.49$) rather than positive ($M = 2.36$, $SD = 1.50$). Stated differently, participants reported feeling more conflicted when their conflicting (i.e., less numerous) reactions

were negative. This result is consistent with our prediction (and inconsistent with the ESM's prediction) and reflects a negativity bias in subjective ambivalence.

Next, we examined whether this valence asymmetry was present at all levels of objective conflict (i.e., attitude mixture), or whether positivity offset might be observable at lower levels of conflict (i.e., when attitudes are less mixed, or univalent; see Cacioppo et al., 1997). As predicted, there was an interaction between conflicting reaction valence and the level of objective conflict, or attitude mixture, $F(2, 15,656) = 9.98$, $p < .001$. This interaction involved a valence asymmetry at high ($b = 0.16$, $t(15,730) = 4.47$, $p < .001$) and moderate ($b = 0.23$, $t(15,764) = 4.00$, $p < .001$) levels of conflict, which disappeared at low levels of conflict ($b = 0.01$, $t(15,030) = 0.17$, $p = .862$; see Table 1). This attenuation at low levels of objective conflict suggested a competing effect due to positivity offset. That is, although negative reactions carry greater weight (because of the negativity bias), at low levels of positive reactions, people still exhibit baseline positivity (i.e., some level of positive output), which then conflicts with their negativity and produces an opposing valence asymmetry.

We also assessed whether these valence asymmetries would affect the point at which maximum subjective ambivalence

Table 2

Study 1: Dependent Measures Correlation Matrix

Dependent measure	Global attitudes	Attitude positivity	Attitude negativity	Objective ambivalence	Subjective ambivalence
Attitude positivity	.73***	—			
Attitude negativity	-.66***	-.64***	—		
Objective ambivalence	.10***	.29***	.07***	—	
Subjective ambivalence	-.18***	-.02*	.15***	.14***	—

Note. Objective ambivalence was calculated from positive and negative reactions using the similarity-difference formula: $(P + N)/2 - |P - N|$.

* $p < .05$. *** $p < .001$.

obtains. Canonical work in ambivalence suggests that maximum ambivalence should be present when individuals have positive and negative reactions of equal magnitude. However, we predict that under these conditions—that is, when people have the exact same number of positive and negative reactions—negativity will overshadow positivity. In fact, as explained earlier, a negativity bias account suggests that greater ambivalence will result from instances in which positive reactions slightly outnumber negative reactions. To test this possibility, we compared subjective ambivalence levels for attitudes composed of equal numbers of positive and negative reactions (i.e., balanced attitudes) with the other high conflict attitudes in the sample.

As predicted, balanced attitudes did not result in maximal ambivalence. In fact, among high conflict attitudes, subjective ambivalence was lower for balanced attitudes ($M = 3.48$, $SD = 1.60$) than for negative conflicting attitudes ($M = 3.61$, $SD = 1.31$; $b = 0.12$, $t(16,007) = 3.47$, $p < .001$), but it did not differ from positive conflicting attitudes ($M = 3.45$, $SD = 1.47$; $b = -0.03$, $t(15,951) = -0.94$, $p = .349$). This finding suggests that when positive and negative reactions are present in equal proportions, negative dominates positive on a psychological level, leading to less subjective ambivalence relative to instances in which positive reactions slightly outnumber negative reactions. Supporting this conclusion, an exploratory analysis treating balanced attitudes as positive conflicting (because of negativity's relatively greater weight) revealed a significant interaction between the valence and number of conflicting reactions ($b = 0.11$, $t(15,880) = 9.63$, $p < .001$). Increasing the number of conflicting reactions increased subjective ambivalence more quickly when those reactions were negative ($b = 0.41$, $t(16,056) = 43.74$, $p < .001$) rather than positive ($b = 0.30$, $t(15,801) = 47.02$, $p < .001$).⁴ That is, the slope for increasing subjective ambivalence was steeper when conflicting reactions were negative rather than positive, reflecting a negativity bias in subjective ambivalence consistent with our predictions but counter to the ESM.

Also important, we obtained evidence for positivity offset that conceptually replicates the findings of Cacioppo et al. (1997). Specifically, the regression line for positive conflicting reactions had a higher y-intercept (1.56) than did the regression line for negative conflicting reactions (1.34), $t(15,335) = 4.87$, $p < .001$. In other words, at extremely low levels of conflicting input (e.g., no positive reactions), negative dominant (or positive conflicting) attitudes were associated with more subjective ambivalence than positive dominant (or negative conflicting) attitudes. This result is consistent with the presence of some baseline positivity in mostly negative attitudes (which results in more subjective ambivalence) that is greater than the presence of baseline negativity in mostly positive attitudes.

Taken together, it appears that ambivalence valence asymmetries are driven in one direction by positivity offset at very low levels of conflict, but in the other direction by negativity bias at higher levels of conflict (i.e., as attitudes become more mixed). Additional data showing global attitudes and subjective ambivalence at every level of positive and negative reactions are included in the supplemental online materials (Tables S1 and S2).

Discussion

Study 1 revealed valence asymmetries in attitude ambivalence that were consistent with the operation of both negativity bias and positivity offset across a wide variety of attitude objects. Specifically, in moderate and high conflict (i.e., more mixed) attitudes, subjective ambivalence was higher when the conflicting reactions were negative. This asymmetry points to a negativity bias. However, in low conflict (i.e., more univalent) attitudes, subjective ambivalence did not differ based on valence, suggesting that positivity offset is more observable at low conflict levels.⁵ In fact, differences in the y-intercept of subjective ambivalence from the regression analysis suggest that positivity offset significantly reversed the negativity bias effect at zero conflicting reactions. These findings depart from the predictions of the ESM, which stipulates that positivity offset and negativity bias are complementary influences that have directionally similar effects on ambivalence. In contrast to this notion, Study 1 suggested that these asymmetries can operate in different directions, producing more ambivalence when people have only negative (vs. only positive) reactions, but less ambivalence when people have mostly negative (vs. mostly positive reactions). Importantly, this study also demonstrated that maximum ambivalence occurs not when positive and negative reactions are present in equal proportion, but rather when positive reactions are more numerous than negative reactions. This finding provided further evidence for the role of negativity bias in shaping ambivalence.

Study 2

Study 2 was designed with several goals in mind. First, we sought to examine the role of negativity bias and positivity offset in subjective ambivalence using an experimental context in which we manipulated the valence and amount of positive and negative information available to participants. We presented all participants with 5 pieces of dominant information about a target individual and varied the amount and valence of conflicting information. Thus, each participant received 5 pieces of positive or negative information, and 0, 1, 2, 3, or 4 pieces of conflicting information of the opposite valence. This design had the benefit of permitting

⁴ Interestingly, a complementary analysis that excludes these balanced attitudes attenuates the interaction to below the level of significance ($p = .224$). This attenuation is consistent with the negativity bias augmentation of balanced attitudes and the above finding that balanced attitudes produced subjective ambivalence levels akin to positive conflicting (but not negative conflicting) attitudes. However, this evidence is merely supportive and not considered theoretically central to our hypotheses. Therefore, analysis of the experimental evidence in Studies 2 and 3 does not rely on this grouping, instead excluding balanced information conditions to provide a more conservative test of the hypothesis.

⁵ As an alternative possibility, it could be that positivity offset appears weaker than negativity bias for self-reported attitudes because when people report low or no positivity, there is little reason to suspect that this masks unreported default positivity *a la* positivity offset. According to this alternative, the attenuation of the valence asymmetry for low conflict attitudes could be due to lack of positivity and negativity at very low levels of conflict. Indeed, absolute levels of ambivalence at this level of conflict were very low. However, such an explanation discounts the intercept difference from the continuous analysis, which supports the operation of positivity offset. Although this issue cannot be resolved in Study 1, the experimental data in Studies 2 and 3 provide more conclusive support for positivity offset in subjective ambivalence.

causal inference with respect to our subjective ambivalence effects. Of importance, it also allowed us to examine participants' positive and negative reactions (and, thus, objective ambivalence) as a dependent variable, which we could not do in Study 1 given that we relied on these reported reactions to create the grouping variable.

In addition, we sought to assess the downstream consequences of our subjective ambivalence effects for other properties of attitude strength. More specifically, we examined attitude certainty and the biasing effect of an attitude on one's thoughts (i.e., thought bias). Although ambivalence and attitude certainty are conceptually and empirically distinct (see Clarkson, Tormala, & Rucker, 2008; Luttrel, Petty, & Briñol, 2016), there is substantial evidence for a negative relationship between these constructs (Bassili, 1996; Jonas, Diehl, & Brömer, 1997; Petrocelli, Tormala, & Rucker, 2007; Rucker, Petty, & Briñol, 2008; Tormala & Rucker, 2007). Thus, we examined whether attitudes held with less ambivalence would also be held with greater certainty. In addition to fostering certainty, it is well-established that strong attitudes exert more impact on thinking than their weaker counterparts—that is, they show a greater tendency to bias thoughts in an attitude-consistent direction (see Krosnick & Petty, 1995). Thus, we expected elevated subjective ambivalence (a classic indicator of weak attitudes) to produce less thought bias. To determine whether subjective ambivalence had implications for attitude certainty and thought bias (i.e., whether more ambivalence would lead to *less* attitude certainty and *less* thought bias) in Study 2, we measured these outcomes directly. We predicted that negativity bias and positivity offset would impact ambivalence as outlined earlier, and that there would be an associated inverse relationship with certainty and thought bias. If true, and if subjective ambivalence mediated these effects, this would offer important evidence that the ambivalence effects are meaningful in the sense that they produce other downstream outcomes.

Method

Participants and procedure. Five hundred sixty online respondents (Female = 42.7%, $M_{\text{Age}} = 34.9$, $SD_{\text{Age}} = 11.9$), recruited using Amazon's Mechanical Turk, participated in exchange for monetary compensation. Each participant was assigned to receive five pieces of information (i.e., traits) about a target person that were either positive or negative. Each participant also received between 0 and 4 conflicting traits of the opposite valence. Thus, in this study design, the amount of dominant information was fixed, whereas the amount of conflicting information varied. A balanced information condition also was included in the design, resulting in a 2 (valence of conflicting traits: positive vs. negative) $\times$ 5 (number of conflicting traits: 0, 1, 2, 3, or 4) + 1 (balanced: 5 positive and 5 negative traits) between participants design. These conditions were designed to generate a range of attitudes from more to less mixed while holding constant the amount of dominant trait information participants had about the target person.

At the outset of the study, participants learned that the purpose of our research was to assess how opinions about people are formed. They read that several people who know (or had known) a specific individual named Frank were interviewed and asked to report the first single-word descriptor that came to mind when

thinking about him. Each word was presented in quotation marks, followed by the name of the (hypothetical) person reporting it. Following this task, participants completed measures of attitudes, ambivalence, and attitude certainty, followed by a thought listing task to assess biased thinking.

Information manipulation. Participants were randomly assigned to one of 11 conditions varying the amount of positive and negative trait information. To generate our stimuli, we drew upon Anderson's (1968) likability ratings for personality traits. Anderson collected likability ratings for 555 unique trait words on a 7-point scale anchored at *least favorable or desirable* (0) and *most favorable or desirable* (6). We calculated an extremity score (absolute value of the raw score minus the scale midpoint [3]) for each word in the original paper. From Anderson's list, we created a set of 42 positive traits and a set of 38 negative traits. Each set contained words with extremity scores between 1.5 and 2.0 in Anderson's original study. We then tested the traits for our study by asking 88 different participants from the same population to rate all 80 words on a 1–7 scale (1 = *bad*, 7 = *good*). From this test, we selected 12 positive and 12 negative items to create trait lists with equivalent extremity (absolute value of the raw score minus the scale midpoint [4]; $M_{\text{pos}} = 2.33$, $SD_{\text{pos}} = 0.60$, $M_{\text{neg}} = 2.29$, $SD_{\text{neg}} = 0.58$; $t(87) = 0.90$, $p = .371$; see Appendix for the full stimulus set). For each participant, traits were randomly drawn without replacement from these lists to create the specific proportion of positive and negative traits for the assigned condition. All assigned traits appeared on a single screen in random order.

Dependent measures.

Attitudes and ambivalence. All measures (global attitudes, positive reactions, negative reactions, and subjective ambivalence) were adapted from those used in Study 1 to reflect changes in the attitude object to a person (Frank). In addition, the global attitude assessment was modified and included three items: "How much do you like Frank?" (1 = *Strongly Dislike*, 9 = *Strongly Like*), "How favorable is your impression of Frank?" (1 = *Not At All Favorable*, 9 = *Very Favorable*), and "How would you describe your feelings toward Frank?" (1 = *Strongly Negative*, 9 = *Strongly Positive*). The attitude positivity and negativity items were used to compute an index of objective ambivalence using the standard similarity-intensity formula popularized by Thompson et al. (1995): $\frac{(P+N)}{2} - |P - N|$. The global attitude ($\alpha = .96$) and subjective ambivalence items (7-point scales; $\alpha = .93$) were internally consistent, so each set was averaged to form its own composite index.

Attitude certainty. Attitude certainty was assessed using a single item adapted from previous research (e.g., Clarkson et al., 2008): "How certain are you of your impression of Frank?" Participants responded on a 1–7 scale (1 = *Not At All Certain*, 7 = *Very Certain*).

Thought listing. To assess thought bias, participants reported their thoughts about Frank by entering them one at a time into a series of text boxes (Cacioppo & Petty, 1981). Once one thought had been entered, a new text box appeared on the screen. Participants were allowed to list as many as 10 thoughts total. After sharing their thoughts, participants were shown each thought they had just entered and were asked to report whether that thought was relevant to the attitude object ("Is this thought about Frank?" 1 = *Yes*, 2 = *No*). Then, after a brief definition of the word *valence*, for

each relevant thought participants were asked to indicate the thought valence ("What valence is this thought?" 1 = *Positive*, 2 = *Neutral*, 3 = *Negative*). An index of thought bias was created for each participant's relevant thoughts by computing the difference between the number of positive and negative thoughts and dividing this difference by the total number of thoughts. To capture the direction of thought bias relative to the attitude manipulation, we then recoded the thought index such that values ranged from -1 to $+1$, with positive values indicating more biased, or skewed, thoughts in the direction of the dominant valence, negative values indicating thoughts skewed in the direction of the conflicting valence, and values close to zero indicating equivalent amounts of positive and negative thoughts. As an indicator of attitude strength, more thought bias would be associated with higher positive scores on this index.

Analysis coding.

Conditions. To examine the interaction between the number and valence of conflicting traits on ambivalence, certainty, and thought bias, we coded each independent variable. The conflicting trait valence code indicated whether the less numerous traits were positive (1) or negative (-1).⁶ Number of conflicting traits was coded with a numerical value ranging from 0 to 4. Although we randomly assigned participants to discrete conditions, we treated the number of conflicting traits as a continuous variable to facilitate analysis by regression. As outlined previously, we predicted an interaction between the valence and number of conflicting traits as well as a simple effect on the intercept (i.e., at zero conflicting traits). The interaction would imply a slope difference consistent with negativity bias and a difference in y intercept consistent with positivity offset.

Positive and negative reactions. Items assessing separate positive and negative reactions were recoded based on assigned trait information condition, resulting in two new variables: dominant reactions and conflicting reactions. The conflicting reactions item was of particular interest because in this study we manipulated the amount of conflicting trait information while holding constant the amount of dominant trait information. We did this because, relative to dominant reactions, conflicting reactions tend to be more closely associated with subjective ambivalence (Priester & Petty, 1996). Thus, regressing conflicting reactions on the number and valence of conflicting traits permits a side-by-side comparison of differences in positive versus negative reactions as a function of the number of conflicting traits presented. Again, an interaction between slopes would be interpreted as evidence for negativity bias whereas differences in the y intercept would provide support for positivity offset.

Results

All cell means and standard deviations are presented in Table 3. A correlation matrix for all dependent measures in Study 2 is included as Table 4.

Positive and negative reactions. We began by regressing recoded positive and negative reactions on the number and valence of conflicting traits. This analysis allowed us to examine whether we obtained the predicted effects of the ESM on attitude valence. As expected, increasing the number of conflicting positive traits increased positive reactions ($b = 0.67$, $t(503) = 7.79$, $p < .001$), while increasing the number of conflicting negative traits in-

creased negative reactions ($b = 0.95$, $t(503) = 10.42$, $p < .001$). Consistent with a negativity bias, however, the slope was steeper for negative rather than positive reactions, as indicated by a significant interaction between valence and number of conflicting traits ($b = 0.28$, $t(503) = 2.26$, $p = .024$). Furthermore, consistent with positivity offset, the y -intercepts for positive and negative reactions differed ($b = 0.97$, $t(503) = 2.09$, $p = .002$). The y intercept for positive reactions was higher when there were zero positive conflicting traits (positive conflicting y -intercept = 2.37) than the y intercept for negative reactions was when there were zero negative conflicting traits (negative conflicting y intercept = 1.40). That is, although negative reactions increased more quickly than positive reactions in response to adding conflicting information of the same valence (negativity bias), minimum positivity was greater than minimum negativity, which was consistent with positivity offset and the original attitude valence findings of Cacioppo et al. (1997).

Objective ambivalence. Following the analysis of separate positive and negative reactions, we submitted the overall objective ambivalence index to analysis. Specifically, we regressed objective ambivalence on the number and valence of conflicting traits. First, number of conflicting traits produced a main effect ($b = 1.02$, $t(503) = 12.62$, $p < .001$) such that objective ambivalence increased as the number of conflicting traits increased. Conflicting trait valence exerted a marginal main effect ($b = -0.41$, $t(503) = -1.80$, $p = .072$), such that negative conflicting traits led to slightly higher objective ambivalence than did positive conflicting traits. However, these main effects were qualified by an interaction, revealing an asymmetry in the rate at which objective ambivalence increased ($b = 0.55$, $t(503) = 3.41$, $p < .001$). As illustrated in Figure 3A, increasing the number of negative conflicting traits ($b = 1.29$, $t(503) = 10.99$, $p < .001$) increased objective ambivalence more quickly than did increasing the number of positive conflicting traits ($b = 0.74$, $t(503) = 6.73$, $p < .001$). That is, the slope was steeper for negative rather than positive conflicting traits.

Importantly, we also observed opposing simple effects at the extreme low and high ends of attitude conflict (i.e., conflicting reactions). First, there was a significant simple effect of conflicting trait valence on the intercept ($b = 1.53$, $t(503) = 3.82$, $p < .001$) such that objective ambivalence was greater at zero positive conflicting traits (positive conflicting y -intercept = 0.86) than at zero negative conflicting traits (negative conflicting y -intercept = -0.67). At maximum conflict, however—that is, at the far right side of Figure 3A—this effect was reversed ($b = 1.21$, $t(503) = 2.39$, $p = .022$). In other words, consistent with positivity offset, univalent negative traits led to

⁶ When the number of conflicting components was equal to the number of dominant components (i.e., 5 positive and 5 negative components), neither valence was numerically dominant (or conflicting) and, therefore, a value of *NA* was used. As a result, the balanced information cell was omitted from the analysis. As described above, although positive and negative information are numerically equivalent in this condition, the negativity bias suggests that negative information likely dominates the positive. Following this assumption, one could argue that when considering conflicting component valence, this cell should be coded as *positive*, rather than *NA*. When using such a coding scheme, all relevant analyses remain significant (or increase in their level of significance), suggesting that the omission of the balanced information cell presents a conservative test of our hypotheses.

Table 3
Study 2: Dependent Measures as a Function of Information Condition

Conflicting valence	Number of conflicting	<i>N</i>	Global attitudes	Attitude positivity	Attitude negativity	Objective ambivalence	Subjective ambivalence	Attitude certainty	Thought bias index
Negative	0	45	7.47 (1.53)	7.40 (1.94)	1.18 (1.87)	-1.29 (2.36)	2.06 (1.11)	5.89 (1.75)	.87 (.21)
Negative	1	37	6.80 (.98)	6.89 (1.93)	2.49 (1.48)	.96 (2.18)	3.21 (1.45)	4.68 (1.45)	.71 (.32)
Negative	2	57	6.03 (.95)	6.07 (1.67)	3.44 (1.73)	2.35 (2.19)	4.42 (1.38)	3.84 (1.54)	.54 (.36)
Negative	3	68	5.29 (1.17)	5.54 (1.69)	4.46 (1.97)	3.53 (2.34)	4.75 (1.60)	3.76 (1.74)	.47 (.35)
Negative	4	48	5.10 (1.22)	5.71 (1.57)	4.92 (2.09)	3.81 (2.48)	5.27 (1.32)	3.27 (1.91)	.49 (.32)
Balanced	5	53	4.82 (1.29)	5.36 (1.69)	4.79 (2.18)	3.43 (2.38)	5.28 (1.52)	3.28 (1.87)	.49 (.38)
Positive	4	57	4.19 (1.48)	4.75 (1.87)	4.93 (2.13)	3.56 (2.48)	5.07 (1.65)	3.47 (2.15)	.53 (.39)
Positive	3	39	3.98 (1.24)	4.67 (2.23)	5.03 (2.58)	2.72 (2.94)	4.37 (1.83)	4.00 (2.05)	.50 (.41)
Positive	2	43	3.93 (1.36)	3.91 (2.06)	5.56 (1.96)	3.10 (2.51)	4.19 (1.53)	3.70 (1.74)	.58 (.43)
Positive	1	61	3.31 (1.55)	3.31 (2.25)	5.95 (1.97)	2.24 (2.92)	3.66 (1.60)	4.13 (1.97)	.66 (.35)
Positive	0	52	3.05 (1.50)	2.00 (2.08)	6.04 (2.36)	.06 (2.54)	3.28 (1.78)	4.52 (2.19)	.73 (.35)

Note. Standard deviations are included parenthetically.

higher objective ambivalence than univalent positive traits. Consistent with negativity bias, when reactions were highly mixed, conflicting negative (or dominant positive) traits led to higher objective ambivalence than conflicting positive (or dominant negative) traits.

Subjective ambivalence. Regressing subjective ambivalence on the number and valence of conflicting traits, we observed similar evidence for both negativity bias (difference of slopes) and positivity offset (difference of intercepts). First, number of conflicting traits had a main effect ($b = 0.61$, $t(503) = 12.56$, $p < .001$) such that subjective ambivalence increased as the number of conflicting traits increased. Interestingly, conflicting trait valence exerted no main effect in this analysis ($b = 0.14$, $t(503) = 1.03$, $p = .305$). However, number of conflicting traits interacted with valence, revealing the predicted asymmetry in the rate at which subjective ambivalence increased ($b = 0.35$, $t(503) = 3.67$, $p < .001$). As illustrated in Figure 3B, increasing the number of negative conflicting traits ($b = 0.79$, $t(503) = 11.14$, $p < .001$) increased subjective ambivalence more quickly than did increasing the number of positive conflicting traits ($b = 0.43$, $t(503) = 6.50$, $p < .001$). There was also a simple effect of conflicting trait valence on the intercept ($b = 0.88$, $t(503) = 3.60$, $p < .001$) such that ambivalence was greater at zero positive conflicting traits (positive conflicting y-intercept = 3.25) than at zero negative conflicting traits (negative conflicting y-intercept = 2.37). This effect reversed at maximum levels of conflicting information ($b = 0.91$, $t(503) = 2.88$, $p = .004$). Again, then, consistent with positivity offset, univalent negative traits led to more ambivalence than univalent positive traits; however, consistent with negativity bias, the slope was steeper for conflicting negative rather than positive traits and the intercept difference reversed at high levels of conflicting information.

Attitude certainty. A similar analysis of the attitude certainty measure also revealed valence asymmetries. Both the valence ($b = 0.31$, $t(503) = 1.85$, $p = .065$) and number ($b = -0.42$, $t(503) = -7.09$, $p < .001$) of conflicting traits exerted main effects on attitude certainty. Notably, though, these main effects were qualified by a significant interaction ($b = 0.38$, $t(503) = 3.23$, $p = .001$). As depicted in Figure 3C, increasing the number of negative conflicting traits decreased certainty more quickly ($b = -0.61$, $t(503) = -7.08$, $p < .001$) than did increasing the number of positive conflicting traits ($b = -0.23$, $t(503) = 2.82$, $p = .005$). In line with positivity offset, there was also a simple

effect of valence on the intercept ($b = 1.09$, $t(503) = 3.70$, $p < .001$) such that participants reported less certainty when they had zero positive conflicting traits (positive conflicting y-intercept = 4.42) rather than zero negative conflicting traits (negative conflicting y-intercept = 5.51). This effect reversed at maximum levels of conflicting information ($b = -0.82$, $t(503) = -2.12$, $p = .035$).

To determine whether the interaction on attitude certainty was mediated by subjective ambivalence, as hypothesized, we conducted a mediated moderation analysis following bootstrapping procedures recommended by Hayes (2013; PROCESS Model 8).⁷ This analysis revealed that subjective ambivalence did predict attitude certainty ($b = -.65$, $t(502) = -14.25$, $p < .001$), and that the indirect effect of the interaction on attitude certainty through subjective ambivalence was significant (95% confidence interval [CI] [-.38 to -.12]). Controlling for objective ambivalence, subjective ambivalence continued to predict attitude certainty ($b = -.63$, $t(501) = -12.84$, $p < .001$), whereas objective ambivalence did not ($b = -.03$, $t(501) = -.86$, $p = .382$), and the indirect effect through subjective ambivalence remained significant (CI [-.27 to -.04]).

Thought bias. We submitted the thought bias index to the same analysis. We expected the results to parallel what we observed with attitude certainty. This analysis revealed a main effect of number of conflicting traits ($b = -0.08$, $t(503) = -7.01$, $p < .001$), but not conflicting valence ($b = 0.00$, $t(503) = 0.05$, $p = .957$). Again, though, we obtained the

⁷ It is worth noting that Hayes (2013) argues against the label “mediated moderation” on the grounds that the product term representing an interaction effect is substantively meaningless, so its indirect effect is essentially meaningless as well. Instead, Hayes suggests viewing all conditional indirect effects as cases of moderated mediation. Importantly, though, there are cases (including the current one) in which it is conceptually meaningful to ask whether an interaction effect on a dependent variable (such as attitude certainty) is driven, at least partly, by the effect of that same interaction on a potential mediator (such as subjective ambivalence). Such an analysis can inform us as to whether the interaction on the dependent variable emerges as a result of the interaction on the mediator. And, in fact, the models used to test mediated moderation and moderated mediation are statistically identical (Hayes, 2013). We use the term “mediated moderation” here because it captures the conceptual relationship of interest—an interaction effect on the mediator, which then directly predicts the dependent measure—but this label does not affect the actual analyses in any way. The analyses themselves follow Hayes’ recommendations (PROCESS Model 8).

Table 4
Study 2: Dependent Measures Correlation Matrix

Dependent measure	Global attitudes	Attitude positivity	Attitude negativity	Objective ambivalence	Subjective ambivalence	Attitude certainty	Thought bias index
Attitude positivity	.73***	—					
Attitude negativity	-.66***	-.49***	—				
Objective ambivalence	-.02	.20***	.39***	—			
Subjective ambivalence	-.01	.11*	.05	.52***	—		
Attitude certainty	.06	-.02	.00	-.34***	-.61***	—	
Thought bias index	-.03	-.12**	.04	-.33***	-.43***	.37***	—

* $p < .05$. ** $p < .01$. *** $p < .001$.

predicted interaction ($b = 0.05$, $t(503) = 2.10$, $p = .036$; see Figure 3D). Consistent with negativity bias, Increasing the number of negative conflicting traits decreased thought bias more quickly ($b = -0.10$, $t(503) = -6.24$, $p < .001$) than did increasing the number of positive conflicting traits ($b = -0.05$, $t(503) = -3.57$, $p < .001$). Furthermore, the y -intercept was

lower when participants received zero positive (positive conflicting y -intercept = 0.71) as opposed to zero negative (negative conflicting y -intercept = 0.81) conflicting traits ($b = 0.10$, $t(503) = 1.76$, $p = .080$). This effect reversed at maximum levels of conflicting information ($b = -0.13$, $t(503) = -1.87$, $p = .062$).

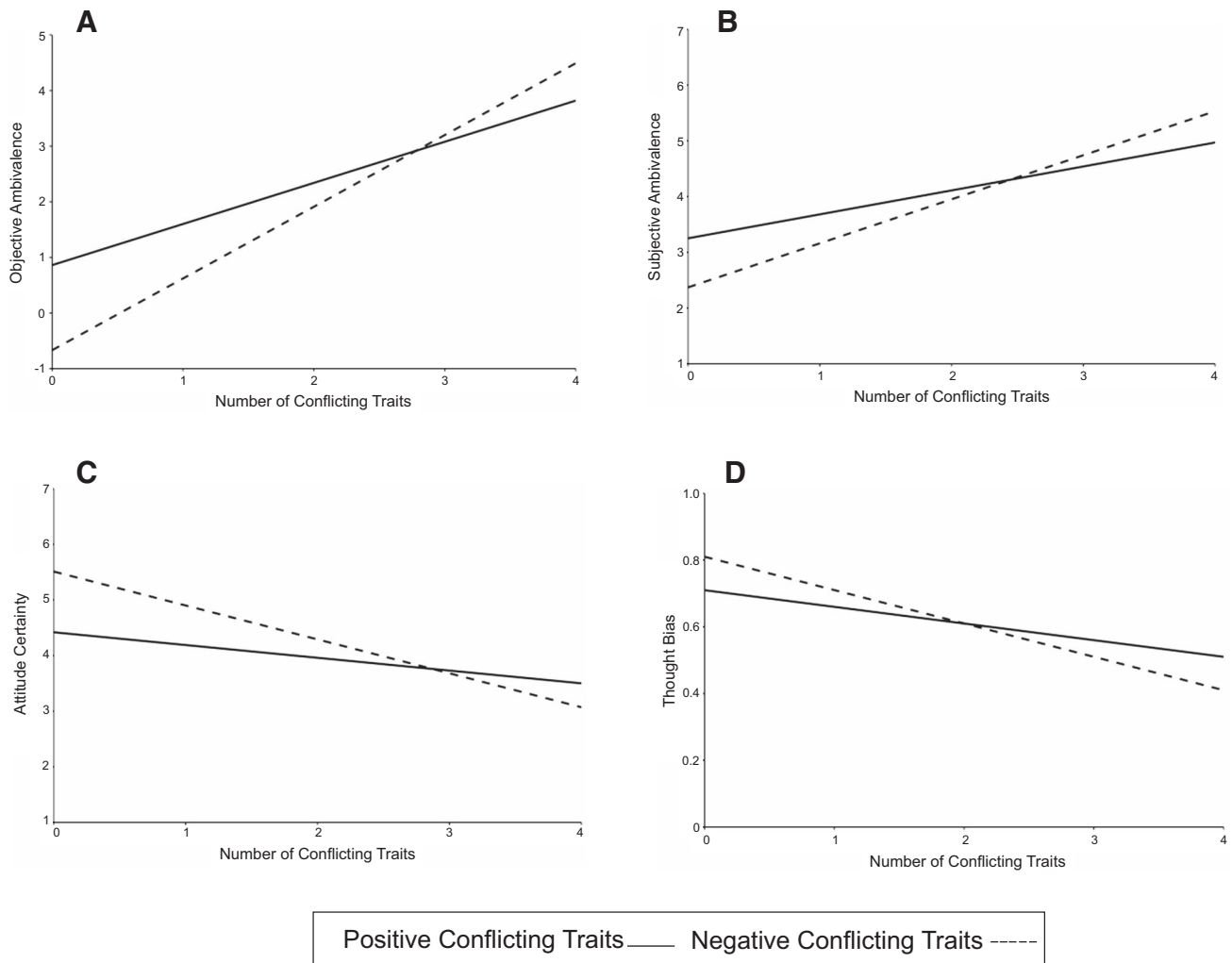


Figure 3. Objective ambivalence (A), subjective ambivalence (B), attitude certainty (C), and thought bias (D) as a function of number and valence of conflicting traits in Study 2.

To determine whether the interaction on thought bias was mediated by subjective ambivalence, we followed the same mediational analysis used to assess attitude certainty. We found that subjective ambivalence did predict thought bias ($b = -.08$, $t(502) = -8.22$, $p < .001$), and that the indirect effect of the interaction on thought bias through subjective ambivalence was significant (CI $[-.05$ to $-.01]$). Controlling for objective ambivalence, both subjective ($b = -.07$, $t(501) = -6.48$, $p < .001$) and objective ($b = -.02$, $t(501) = -3.14$, $p < .01$) ambivalence predicted thought bias, and the indirect effect on thought bias through subjective ambivalence remained significant (CI $[-.03$ to $-.004]$).

Supplemental analyses. Finally, we conducted a series of follow-up analyses testing the effect of positive versus negative valence at each level of conflicting reactions for global attitudes, objective ambivalence, subjective ambivalence, attitude certainty, and thought bias. The results of these analyses are provided in the supplemental online materials (Tables S3–S7).

Discussion

Study 2 provided experimental evidence of opposing roles of positivity offset and negativity bias in subjective ambivalence that were consistent with the findings from Study 1. Interestingly, although we focused our primary hypotheses on outcomes pertaining to subjective ambivalence, we observed similar results on measures of positivity and negativity and objective ambivalence. These findings suggest that positivity offset and negativity bias impacted participants' feelings of positivity and negativity as well as their overall subjective conflict. This insight is generally compatible with our conceptualization, but it does highlight a potential alternative explanation that is worth addressing. Specifically, could it be that the negative traits were seen as more negative than the positive traits were positive? If so, our ambivalence findings could stem from poorly calibrated stimuli rather than generalizable differences in people's underlying psychological reactions to positive and negative information.

We have two responses to this issue. First, we pretested the traits used in this study and found no differences in the perceived extremity of the positive and negative trait terms. In fact, if anything, the positive traits were seen as slightly more extreme than the negative traits. Second, we reanalyzed the subjective ambivalence data controlling for objective ambivalence. Although objective ambivalence was a significant predictor of subjective ambivalence ($b = 0.23$, $t(502) = 9.22$, $p < .001$), as it has been in past research, the interaction between valence and number of conflicting reactions on subjective ambivalence remained significant ($b = 0.23$, $t(502) = 2.54$, $p = .011$). This suggests that the observed asymmetries in subjective ambivalence effects were not a mere function of differential perceptions of how extreme the traits were perceived to be. That subjective ambivalence mediated the attitude certainty and thought bias effects controlling for objective ambivalence further bolsters this conclusion.

Finally, this study provided evidence for valence asymmetries in other related attitude strength outcomes: attitude certainty and thought bias. On each of these metrics, we observed an interaction between the valence and amount of conflicting information, and we found that these interactions were mediated by subjective ambivalence. This result is consistent with the notion that the

subjective ambivalence effects we observed had meaningful downstream consequences for other dimensions of attitude strength.

Study 3

The purpose of Study 3 was twofold. First, we sought to replicate Study 2 using different proportions of positive and negative information. Second, we aimed to examine the prediction that maximum ambivalence occurs at a nonbalanced proportion of positive and negative information (i.e., that the ambivalence distribution is skewed). These goals were accomplished by varying the proportion of positive versus negative traits while holding constant the total number of traits. This allowed us to observe how ambivalence varies along a single reciprocal activation diagonal (i.e., from the top left to the bottom right of Figure 2D; perpendicular to the coactivity diagonal). Examining attitudes along a reciprocal activation continuum such as this is useful because it spans a series of attitudes based on the same total amount of information where the only variable aspect is the relative proportion of positive to negative information (for further description, see Cacioppo et al., 1997). In essence, this allows us to compare the obtained results from Study 3 to the hypothesized density map displayed in Figure 2D.

Therefore, each participant received 12 total pieces of information in varying proportions of positive and negative. As such, in contrast to Study 2 (where amount of dominant information was fixed), in this study the total amount of information was fixed while the proportion of dominant and conflicting traits varied in a reciprocal fashion. Participants were assigned to one of 13 conditions encompassing the full possible range of attitude compositions from 100% positive to 100% negative. These 13 conditions were used to create a 2 (conflicting valence: positive vs. negative) $\times$ 6 (number of conflicting traits: 0, 1, 2, 3, 4, or 5) + 1 (balanced: 6 positive traits, 6 negative traits) between participants design.

This experimental design allowed us to assess where ambivalence peaks as the proportion of positive to negative information varies. That is, it allows us to answer: What is the precise proportion of positive and negative information at which ambivalence maximizes? If ambivalence rises more steeply when negative conflicting information is added to mostly positive information (as opposed to vice versa), it should take less negative conflicting information to produce maximum ambivalence. Again, although the effect of positivity offset differs directionally from that of negativity bias, this should manifest only when attitudes are less mixed (e.g., when people have univalent information). Because ambivalence increases with the coactivation of positivity and negativity, subjective ambivalence should maximize when trait information is heavily mixed. Therefore, finding that subjective ambivalence maximizes (and attitude strength minimizes) when positive traits slightly outnumber negative traits (e.g., when people receive 7 positive and 5 negative traits) would constitute further evidence for the role of negativity bias in shaping subjective ambivalence.

Method

Participants and procedure. Six hundred fifty-four online respondents (Female = 57.1%, $M_{\text{Age}} = 32.1$, $SD_{\text{Age}} = 10.4$) recruited using Amazon's Mechanical Turk participated in exchange for monetary compensation. The procedure was very sim-

Table 5

Study 3: Dependent Measures as a Function of Information Condition

Information proportion	<i>N</i>	Global attitudes	Attitude positivity	Attitude negativity	Objective ambivalence	Subjective ambivalence	Attitude certainty	Thought bias index
12P/0N	51	5.95 (1.10)	7.35 (1.29)	1.10 (1.82)	-1.23 (2.53)	2.16 (1.37)	5.20 (1.72)	.80 (.42)
11P/1N	49	5.59 (.76)	7.00 (1.67)	2.22 (1.84)	.84 (2.83)	2.93 (1.33)	4.90 (1.54)	.71 (.39)
10P/2N	51	5.10 (.94)	6.20 (1.87)	3.04 (1.98)	1.60 (2.47)	3.33 (1.56)	4.84 (1.46)	.60 (.44)
9P/3N	50	4.74 (.79)	6.20 (1.82)	3.56 (2.07)	2.44 (2.60)	3.93 (1.59)	4.30 (1.66)	.53 (.48)
8P/4N	47	4.16 (.95)	5.91 (1.63)	4.57 (2.07)	3.59 (2.42)	4.70 (1.60)	3.77 (1.66)	.31 (.47)
7P/5N	50	4.20 (.86)	5.66 (1.75)	4.40 (2.16)	3.37 (2.41)	5.31 (1.46)	3.34 (1.56)	.24 (.46)
6P/6N	50	3.51 (.94)	5.14 (1.68)	5.34 (2.12)	3.92 (2.55)	5.03 (1.60)	3.42 (1.80)	.37 (.48)
5P/7N	48	3.65 (1.03)	5.15 (2.02)	5.06 (2.09)	3.48 (2.57)	4.74 (1.44)	3.50 (1.69)	.44 (.48)
4P/8N	52	2.91 (1.03)	3.39 (1.78)	5.90 (1.88)	2.62 (2.33)	3.99 (1.67)	3.92 (1.71)	.47 (.48)
3P/9N	53	2.84 (1.17)	3.60 (2.35)	6.02 (2.44)	1.77 (2.61)	3.81 (1.65)	4.09 (1.75)	.62 (.43)
2P/10N	49	2.61 (.82)	3.02 (1.87)	5.90 (1.92)	1.40 (2.04)	3.68 (1.50)	4.04 (1.83)	.60 (.53)
1P/11N	52	2.67 (1.21)	2.83 (2.18)	5.90 (2.10)	1.56 (2.75)	3.09 (1.45)	4.48 (1.79)	.67 (.50)
0P/12N	52	2.43 (1.29)	2.09 (2.23)	6.06 (2.40)	.15 (2.36)	3.14 (1.72)	4.54 (2.01)	.71 (.50)

Note. P = positive traits. N = negative traits. Standard deviations are included parenthetically. Bold indicates apical proportion of information for each index.

ilar to that of Study 2, the exception being that participants were assigned to one of 13 experimental conditions constituting the full range of combinations of positive and negative traits. After the manipulation, participants completed the same measures used in Study 2.

Information manipulation. All participants in Study 3 were exposed to the same total number of traits, varying only whether each trait was of positive or negative valence. The traits came from the same list used in Study 2, but participants received any possible combination of positive and negative traits along the continuum from 12 (100%) positive to 12 (100%) negative (see Table 5 for a complete list).

Dependent measures. The attitude ($\alpha = .96$), positive reactions, negative reactions, subjective ambivalence ($\alpha = .92$), attitude certainty, and thought bias measures and all codes and measure transformations were adopted directly from Study 2. The one exception was that our attitude measures were adapted to use 7-point scales in Study 3.

Results

All cell means and standard deviations are presented in Table 5. A correlation matrix for all dependent measures in Study 3 is included as Table 6.

Positive and negative reactions. As in Study 2, we first examined positive and negative reactions to determine if we obtained the attitude valence effects predicted by the ESM. Specifically, positive and negative reactions (recoded as a single conflicting reactions item) were regressed on the number and valence of conflicting traits. Adding positive conflicting traits increased positive reactions ($b = 0.49$, $t(600) = 7.14$, $p < .001$), while adding negative conflicting traits increased negative reactions ($b = 0.69$, $t(600) = 9.87$, $p < .001$). Consistent with negativity bias, however, attitude negativity increased faster than did attitude positivity, as indicated by a significant interaction between valence and number of conflicting traits ($b = 0.19$, $t(600) = 1.97$, $p = .049$). In line with positivity offset, at zero conflicting traits, positive reactions (positive conflicting y-intercept = 2.10) were greater than negative reactions (negative conflicting y-intercept = 1.47; $b = 0.68$, $t(600) = 2.30$, $p = .022$).

Objective ambivalence. Regressing objective ambivalence on the number and valence of conflicting traits revealed a main effect of number of conflicting traits ($b = 0.75$, $t(600) = 12.40$, $p < .001$), such that objective ambivalence increased as the number of conflicting traits increased. Conflicting trait valence exerted no main effect ($b = -0.02$, $t(600) = -0.08$, $p = .941$). However, replicating Study 2, we found a significant interaction between the number and valence of conflicting traits ($b = 0.34$, $t(600) = 2.84$, $p = .005$). Again, this interaction indicated differential slopes: Adding conflicting traits increased objective ambivalence more quickly when those traits were negative ($b = 0.92$, $t(600) = 10.74$, $p < .001$) rather than positive ($b = 0.57$, $t(600) = 6.78$, $p < .001$; see Figure 4A). Consistent with positivity offset, the intercepts differed such that receiving zero negative conflicting traits led to less objective ambivalence (negative conflicting y-intercept = -0.53) than zero positive conflicting traits (positive conflicting y-intercept = 0.39 ; $b = 0.92$, $t(600) = 2.55$, $p = .011$). Consistent with negativity bias, this effect reversed at maximum levels of conflicting information ($b = 1.13$, $t(600) = 2.40$, $p = .017$). Interestingly, as indicated in Table 5 and consistent with a traditional valence-agnostic perspective on ambivalence, objective ambivalence maximized when participants were exposed to balanced information (6 positive and 6 negative traits; $M = 3.92$, $SD = 2.55$; see also Figure 5A).

Subjective ambivalence. The same analysis on subjective ambivalence revealed a main effect of number of conflicting traits ($b = 0.46$, $t(600) = 12.67$, $p < .001$), such that subjective ambivalence increased as the number of conflicting traits increased. There was no main effect of conflicting trait valence ($b = -0.06$, $t(600) = -0.49$, $p = .625$). Again, though, we observed a significant interaction between the number and valence of conflicting traits ($b = 0.31$, $t(600) = 4.27$, $p < .001$). As illustrated in Figure 4B, adding conflicting traits increased ambivalence more quickly when those traits were negative ($b = 0.62$, $t(600) = 11.94$, $p < .001$) rather than positive ($b = 0.31$, $t(600) = 5.96$, $p < .001$). Furthermore, there was an intercept difference such that receiving zero negative conflicting traits led to less ambivalence (negative conflicting y-intercept = 2.18) than zero positive conflicting traits (positive conflicting y-intercept = 2.97 ;

Table 6
Study 3: Dependent Measures Correlation Matrix

Dependent measure	Global attitudes	Attitude positivity	Attitude negativity	Objective ambivalence	Subjective ambivalence	Attitude certainty	Thought bias index
Attitude positivity	.79***	—					
Attitude negativity	-.69***	-.52***	—				
Objective ambivalence	-.04	.18***	.41***	—			
Subjective ambivalence	-.03	.11**	.15***	.60***	—		
Attitude certainty	.08*	.02	-.04	-.39***	-.67***	—	
Thought bias index	-.08*	-.09*	-.01	-.43***	-.55***	.46***	—

* $p < .05$. ** $p < .01$. *** $p < .001$.

$b = 0.79$, $t(600) = 3.62$, $p < .001$). This effect reversed at maximum levels of conflicting information ($b = 1.08$, $t(600) = 2.77$, $p < .001$). Finally, as indicated in Table 5 and inconsistent with a valence-agnostic perspective on ambivalence, subjective ambivalence maximized when participants were exposed to slightly more positive traits than negative traits (7 positive and 5 negative traits; $M = 5.31$, $SD = 1.46$; see also Figure 5B).

Attitude certainty. Conducting the same analysis on attitude certainty revealed a main effect of number of conflicting traits (more conflicting traits led to less attitude certainty; $b = -0.29$, $t(600) = -7.04$, $p < .001$), and a marginal effect of conflicting trait valence ($b = 0.25$, $t(600) = 1.77$, $p = .078$) suggesting that negative conflicting traits led to less attitude certainty than did positive conflicting traits (or that negative dominant information

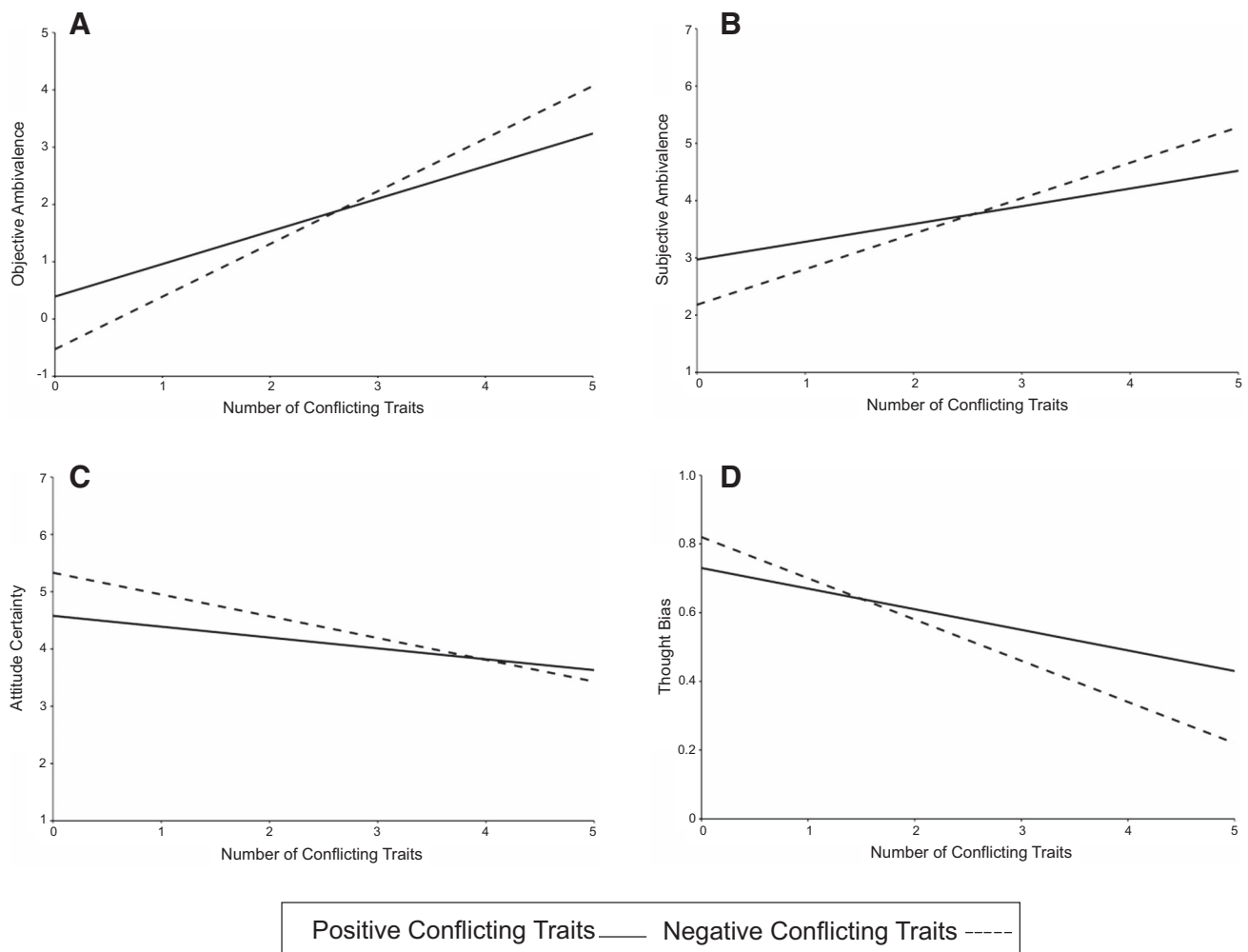


Figure 4. Objective ambivalence (A), subjective ambivalence (B), attitude certainty (C), and thought bias (D) as a function of number and valence of conflicting traits in Study 3.

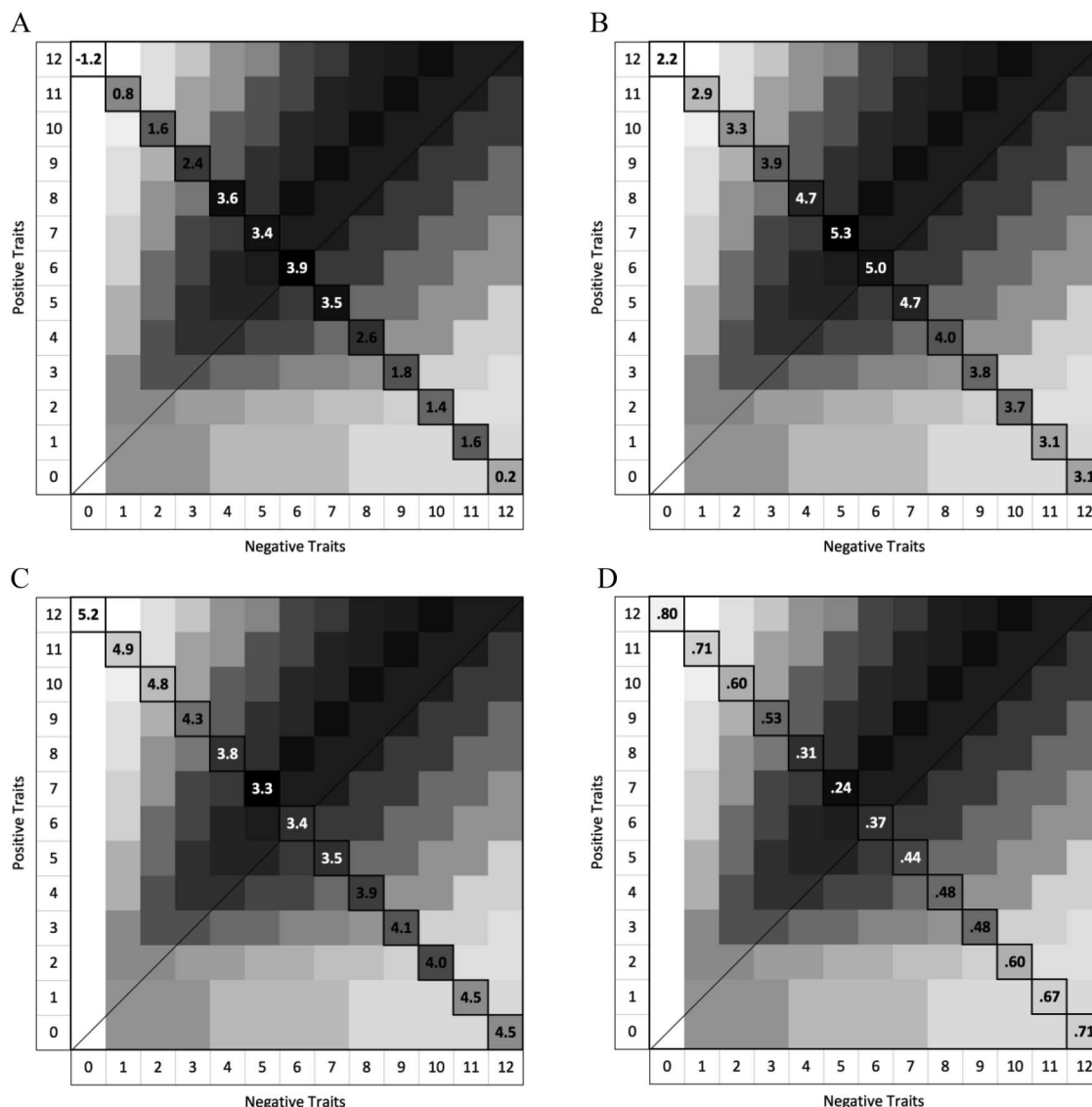


Figure 5. Density plots of objective ambivalence (A), subjective ambivalence (B), attitude certainty (C), and thought bias (D) with cell means overlaid along a reciprocal continuum of mixed information. Black and gray cells reflect hypothesized responses based on the impact of negativity bias and positivity offset (darker shades = more objective and subjective ambivalence; less certainty and thought bias), also presented in Figure 2D. Numbers indicate the actual cell means from Study 3.

produced more certainty than did positive dominant information; see also Bizer et al., 2011). Most importantly, these main effects were qualified by a significant interaction between number and valence of conflicting traits ($b = -0.18$, $t(600) = -2.26$, $p = .024$). As illustrated in Figure 4C, mirroring the ambivalence results, increasing the number of conflicting traits reduced attitude certainty more quickly when those traits were negative ($b = -0.38$, $t(600) = -6.56$, $p < .001$) rather than positive ($b = -0.19$, $t(600) = -3.39$, $p < .001$). Additionally, there was an intercept difference such that zero negative conflicting traits led to greater certainty (y-intercept = 5.33) than zero positive conflicting traits (y-intercept = 4.58; $b = 0.75$, $t(600) = 3.08$, $p = .002$). This effect showed a nonsignificant tendency to reverse at maximum levels of conflicting information

($b = -0.35$, $t(600) = -1.11$, $p = .269$). Finally, as shown in Table 5, attitude certainty minimized when positive traits slightly outnumbered negative traits (7 positive and 5 negative traits; $M = 3.34$, $SD = 1.56$; see also Figure 5C).

To assess whether the interaction on attitude certainty was mediated by subjective ambivalence, we followed the same procedure as in Study 2. This analysis revealed that subjective ambivalence predicted certainty ($b = -0.70$, $t(599) = -19.98$, $p < .001$), and that the indirect effect of the interaction on certainty through subjective ambivalence was significant (CI $[-.33$ to $-.13]$). Controlling for objective ambivalence, subjective ambivalence continued to predict certainty ($b = -0.73$, $t(598) = -17.89$, $p < .001$), whereas objective ambivalence did not ($b = 0.03$,

$t(598) = 1.07, p = .284$), and the indirect effect through subjective ambivalence remained significant (CI $[-.25$ to $-.06]$).

Thought bias. Finally, we examined the thought bias index. Here we found main effects for both the number ($b = 0.11, t(600) = 9.26, p < .001$) and valence ($b = -0.06, t(600) = -6.02, p = .003$) of conflicting traits. Most important, there was a significant interaction between these variables ($b = 0.06, t(600) = 2.67, p = .007$; see Figure 4D): Adding conflicting traits reduced thought bias more quickly when those traits were negative ($b = -0.12, t(600) = -7.39, p < .001$) rather than positive ($b = -0.06, t(600) = -3.62, p < .001$). Although not statistically significant, there also was a directional intercept difference consistent with positivity offset such that zero negative conflicting traits (y -intercept = 0.82) led to more biased thinking than zero positive conflicting traits (y -intercept = 0.73; $b = 0.09, t(600) = 1.41, p = .158$). This effect significantly reversed at maximum levels of conflicting information ($b = -0.26, t(600) = -3.04, p = .002$). Finally, paralleling attitude certainty, thought bias minimized when positive traits were slightly more numerous than negative traits (7 positive and 5 negative traits; $M = 0.37, SD = 0.48$, see Table 5 and Figure 5D).

To determine whether the interaction on thought bias was mediated by subjective ambivalence, we followed the same mediational analysis as above. We found that subjective ambivalence predicted thought bias ($b = -0.14, t(599) = -13.22, p < .001$), and that the indirect effect of the interaction on thought bias through subjective ambivalence was significant (CI $[-.07$ to $-.02]$). Controlling for objective ambivalence, both subjective ($b = -0.12, t(598) = -10.02, p < .001$) and objective ($b = -0.02, t(598) = -3.14, p < .01$) ambivalence predicted thought bias, and the indirect effect on thought bias through subjective ambivalence remained significant (CI $[-.04$ to $-.01]$).

Supplemental analyses. Finally, as in Study 2, we conducted a series of follow-up analyses testing the effect of positive versus negative valence at each level of conflicting reactions for global attitudes, objective ambivalence, subjective ambivalence, attitude certainty, and thought bias. The results of these analyses are provided in the supplemental online materials (Tables S8–S12).

Discussion

Study 3 took a unique approach to exploring valence asymmetries in attitude ambivalence by observing how ambivalence varied along a reciprocal continuum of mixed information. In addition to replicating the effect of negativity bias and positivity offset for reciprocal proportions of information, this study examined our prediction that subjective ambivalence does not maximize at equal proportions of positive and negative information. In Study 3, subjective ambivalence maximized at a point along the continuum at which positive information outnumbered negative information. In fact, in a post hoc analysis, we compared each mixed information cell in Study 3 with its two adjacent cells (contrast: $-1, 2, -1$; where the cell of interest takes the middle position). Consistent with our hypothesis (and visual inspection of the means), the only mixed information cell identified as having peak, or maximum, subjective ambivalence relative to its surrounding cells was the 7 positive, 5 negative cell, $t(641) = 1.67, p = .048$, one-tailed. Also interesting, the balanced information cell (6 positive, 6 negative)—which classic models of ambivalence predict should produce max-

imum conflict—did not produce higher subjective ambivalence than adjacent cells, $t(641) = -0.20, p = .492$, one-tailed. Similar patterns were obtained with attitude certainty and thought bias.

This result fits our predictions based on negativity bias, but it is also important in revealing a violation of any prior formula of ambivalence that overlooks valence asymmetries. Indeed, all prior formulae that have been proposed to estimate ambivalence from the amount of positive and negative information (or reactions) people have directly imply that maximum conflict occurs when positive and negative information are present in equal proportions. Our findings for subjective ambivalence contradict this assumption and suggest that such formulae should account for valence asymmetries. Based on our findings with respect to attitude certainty and thought bias, the differences in subjective ambivalence appear to be real and meaningful, having downstream implications for other aspects of attitude strength.

Finally, although we used the same stimuli as in Study 2, which we had pretested to ensure that they were equally extreme across valence, we again reanalyzed the subjective ambivalence data controlling for objective ambivalence. Replicating Study 2, although objective and subjective ambivalence were correlated ($b = 0.30, t(599) = 13.94, p < .001$), the interaction effect on subjective ambivalence remained significant ($b = 0.21, t(599) = 3.27, p = .001$). Combined with the mediation analyses of the attitude certainty and thought bias results, this result further suggests that subjective ambivalence asymmetries were not a mere function of differential perceptions of positive and negative trait extremity.

General Discussion

The present research was designed to more precisely describe how mixed reactions or information combine to shape subjective ambivalence. Identifying the specific conditions that produce subjective ambivalence has great theoretical import because ambivalence is an integral aspect of attitude strength (Bassili, 1996; Krosnick & Petty, 1995). As a result, documenting how and when ambivalence manifests from conflicting information improves the prediction of people's thoughts, feelings, and actions (Conner & Armitage, 2008). Although researchers have proposed and validated numerous formulae to predict ambivalence in past work (e.g., Priester & Petty, 1996; Thompson et al., 1995; Riketta, 2000), not a single currently available formulation accounts for the differential impact of positive versus negative valence (e.g., Priester & Petty, 1996; Thompson et al., 1995), despite considerable evidence for valence asymmetries in attitudes and social cognition (Bizer & Petty, 2005; Cacioppo et al., 1997; Ito et al., 1998; Fazio et al., 2004; Kahneman & Tversky, 1979; Pietri et al., 2013; Rozin & Royzman, 2001).

Our work helps merge these literatures and, thus, create more nuanced predictions about how variable proportions of positivity and negativity foster ambivalence. Across three studies, we found consistent support for valence asymmetries—in particular, negativity bias and positivity offset. These asymmetries emerged using both experimental and correlational methods, and we observed them across a broad range of topics (see Study 1). In line with negativity bias, we found that adding conflicting information (or reactions) had a greater effect on subjective ambivalence when that conflicting information was negative rather than positive. Especially at high levels of objective attitude conflict, this asymmetry

meant that people tended to feel more ambivalent when their conflicting information or reactions were negative as opposed to positive. Interestingly, as objective conflict was reduced (i.e., attitudes became less mixed), this valence asymmetry attenuated, and ultimately reversed when people had zero conflicting information, consistent with positivity offset. One intriguing consequence of these asymmetries was that maximum ambivalence occurred when positive information actually outnumbered negative information, rather than when they were in present in equal proportions. Also noteworthy, the observed variation in ambivalence fostered changes in other aspects of attitude strength. That is, elevated subjective ambivalence led to reduced attitude certainty and less biased thinking in Studies 2 and 3—classic indications of reduced attitude strength.

Of course, we also calculated objective ambivalence in Studies 2 and 3. We found that objective ambivalence also displayed a similar interaction to subjective ambivalence, revealing differential slopes and intercepts consistent with negativity bias and positivity offset, respectively. However, the subjective ambivalence effects remained significant when we controlled for these effects. Moreover, as shown in Table 5, the objective ambivalence maximization point deviated from the maximization (minimization) point for subjective ambivalence (certainty and thought bias). Indeed, objective ambivalence maximized when positive and negative information were present in equal proportions—not when positive dominated slightly—suggesting that this calculation was more tied to the information people received and did not fully map onto participants' subjective experience of conflict. In fact, the imperfect relation between objective and subjective ambivalence measures is an important and frequently discussed matter in the broader ambivalence literature (e.g., DeMarree et al., 2014; Priester & Petty, 2001). Future work should consider how to integrate the concept of negativity bias and positivity offset into the estimation of objective ambivalence.

Importantly, the results from the current work reach beyond the predictions and evidence supplied by Cacioppo et al. (1997) in their work on the ESM. In fact, our subjective ambivalence findings are at odds with the predictions of that model. As discussed earlier, Cacioppo et al. predicted that the activation and proliferation of negative information or reactions leads to greater ambivalence. This prediction suggested that (a) ambivalence would be higher when people had univalent negative rather than univalent positive information, and (b) ambivalence would be higher when people had dominant negative rather than dominant positive information. The logic was that positivity offset and negativity bias have directionally similar effects.

The current findings were compatible with the ESM predictions for positivity offset when people have univalent information or reactions, but we predicted and found that negativity bias produced a valence asymmetry in the opposite direction as information or reactions became more mixed. In essence, whereas the ESM makes a main effect prediction regarding the impact of valence on ambivalence, we predicted and found an interaction pattern. Positivity offset produces more ambivalence when people have univalent negative rather than positive reactions, but the steeper rise in ambivalence as negative (vs. positive) conflicting reactions are added ultimately reverses the pattern such that people feel more ambivalent when they have conflicting negative (or dominant positive) reactions. We were able to uncover this pattern by in-

cluding a wider spectrum of conditions (ranging from univalent to fully balanced) that allowed us to tease apart the effects of positivity offset and negativity bias. One key contribution from the current studies, then, comes from the recognition that negativity bias and positivity offset can have divergent effects on subjective feelings of attitude conflict.

In short, the current research departs from the ESM in the following ways: (a) it reveals that negativity bias and positivity offset do not necessarily work in concert in shaping subjective ambivalence, (b) it suggests that attitude mixture (i.e., degree of conflict) accounts for which asymmetry is more readily observed, and (c) it establishes that these patterns manifest on other, related dimensions of attitude strength. The ESM made groundbreaking contributions in outlining asymmetries in attitude valence, but it was ambiguous with respect to how these asymmetries affect ambivalence. The current work sought to address this ambiguity.

Remaining Questions and Additional Evidence

There are a number of questions that could be raised with respect to our findings. As one example, would two individuals who have the exact same amount of dominant and conflicting information differ in subjective ambivalence if the valence of their information varies? In the introduction, we discussed a hypothetical example in which someone has 8 positive and 4 negative reactions or 4 positive and 8 negative reactions to the same stimulus, and in contrast to the ESM we predicted that the former individual would feel more ambivalent than the latter. What would the evidence show? As another question, is it possible that the current effects stem from something specific about the trait stimuli used in Studies 2 and 3? For example, perhaps our negative traits differed from our positive traits in some way other than extremity. Would our experimental findings replicate outside the domain of person perception? Perhaps negativity bias and positivity offset are especially likely to emerge in the context of impression formation.

To address these issues, we conducted two follow-up experiments. In the first ($N = 233$; Female = 57.1%, $M_{\text{Age}} = 22.7$, $SD_{\text{Age}} = 5.3$), we randomly assigned participants to receive negative conflicting (8 positive, 4 negative), positive conflicting (8 negative, 4 positive), or balanced (6 positive, 6 negative) trait information about a target person. To examine whether our earlier findings were limited to the particular traits used as stimuli in Studies 2 and 3, we generated a new list of traits using antonyms (see Appendix) to create positive and negative traits of approximately equal extremity. As it turns out, a test of the extremity of these 12 antonym pairs revealed that the positive traits were actually more positive than the negative were negative. Although unintentional, using more extreme positive traits makes this a conservative test of the role of negativity bias in subjective ambivalence. Despite the conservative nature of the test, a one-way analysis of variance (ANOVA) revealed that subjective ambivalence ($\alpha = .86$) was lower in the positive conflicting condition ($M = 4.26$, $SD = 1.46$) than in the negative conflicting ($M = 5.20$, $SD = 1.25$; $F(1, 230) = -8.64$, $p < .001$) or balanced conditions ($M = 5.29$, $SD = 1.31$, $F(1, 230) = -9.57$, $p < .001$), which did not differ, $F(1, 230) = 0.80$, $p = .689$. This result was consistent with a negativity bias account and suggests that the findings of the current studies are not constrained to the particular traits we employed in Studies 2 and 3.

To further establish the robustness of our findings, we also tested the valence asymmetry effect outside the domain of person perception. Although Study 1 provided correlational evidence for the current hypothesis using a series of other issues, experimental evidence in a new domain would provide additional confidence in the generalizability of the observed effects. To this end, we developed materials for a hypothetical computer tablet and asked participants to take part in a product evaluation task. The materials employed positive and negative descriptions of the same product dimensions or features (see Appendix). Expanding on the design, we also included two univalent conditions (positive only or negative only). Thus, participants were randomly assigned to one of five conditions: positive only (12 positive), negative only (12 negative), positive conflicting (8 positive, 4 negative), negative conflicting (8 positive, 4 negative), or balanced (6 positive, 6 negative). All participants received 12 pieces of information—specifically, short 1- to 4-word reviews of the tablet, each highlighting a single attribute of the tablet itself. Participants viewed either a positive or a negative description of each attribute dimension. For example, on the dimension of weight, participants saw “heavy” (negative) or “lightweight” (positive). On the dimension of durability, participants saw “durable” (positive) or “flimsy” (negative).

The subjective ambivalence composite ($\alpha = .94$) for this sample ($N = 250$; Female = 48.0%, $M_{\text{Age}} = 35.0$, $SD_{\text{Age}} = 11.7$) was submitted to a one-way ANOVA, $F(4, 245) = 25.51$, $p < .001$, which provided support for negativity bias in subjective ambivalence. Ambivalence was lower in the positive conflicting condition ($M = 3.96$, $SD = 1.73$) than in the negative conflicting ($M = 4.88$, $SD = 1.63$; $F(1, 245) = -5.77$, $p = .004$) or balanced conditions ($M = 4.96$, $SD = 1.60$, $F(1, 245) = -6.22$, $p = .002$), which did not differ from one another, $F(1, 245) = 0.52$, $p = .797$. As expected, univalent information produced lower ambivalence than did the mixed information conditions, $F(1, 245) = -18.66$, $p < .001$. However, there was no difference between the positive only ($M = 2.88$, $SD = 1.51$) and negative only ($M = 2.46$, $SD = 1.53$) conditions to indicate positivity offset in these data, $F(1, 245) = 2.60$, $p = .196$, suggesting that perhaps positivity offset might be less robust than negativity bias in this context (see also Sears, 1983).

Collectively, these supplementary studies help establish the generality of valence asymmetry in subjective ambivalence. Indeed, taken together, we have found evidence of asymmetry in a correlational study across multiple attitude issues, in person perception experiments using conflicting traits, and in a product perception context in which participants evaluated a computer tablet on a variety of attributes. In each case we found evidence of robust valence asymmetry in ambivalence. In future work, it would be useful to explore the boundaries on these effects.

Ambivalence and Thought Bias

Another issue worthy of attention in future research involves the relationship between attitude ambivalence and thought bias. In Studies 2 and 3 of the current work, we predicted and found that participants would show more biased thinking (i.e., a greater proportion of attitude-consistent thoughts) when they felt less subjectively ambivalent. This was an attitude strength prediction: Strong attitudes are known to exert more biasing influence over people's thoughts, and ambivalence is a classic dimension of (low) attitude strength. However, there is work suggesting that people sometimes show more biased processing when they feel more rather than less ambivalent,

because biased processing offers a means of reducing ambivalence. This has been shown mostly with respect to what information people prefer or process in a message (Bell & Esses, 2002; Clark, Wegener, & Fabrigar, 2008), but there is some evidence for it in thought listing tasks similar to those employed in Studies 2 and 3 (Nordgren, van Harreveld, & van der Pligt, 2006).

Future research exploring moderators of the ambivalence-bias relationship would help clarify the conditions under which ambivalence increases versus decreases biased thinking. It seems reasonable to surmise that in Studies 2 and 3 of the current work, which focused on traits associated with an unfamiliar target person, the need to resolve ambivalence might have been less salient or pronounced than in some past studies (e.g., in Nordgren et al., 2006, participants were induced to feel ambivalent about genetically modified food). In fact, Nordgren et al. found that ambivalent individuals only displayed biased thinking when they felt tense and attributed that feeling to their ambivalence. Perhaps participants in our studies were less discomforted by their ambivalence. There may even be some domain specificity to these findings. Perhaps some attitude objects (e.g., a food one might eat) are more likely than others (e.g., an unfamiliar target person) to provoke a need to resolve ambivalence. If so, they might alter the relationship between ambivalence and biased thinking. Resolving this issue is beyond the scope of the current research, but future work delving into the ambivalence-bias link would be useful.

Objective and Subjective Ambivalence Correlation

Finally, although subjective ambivalence continues to be the gold standard for assessing how conflicted someone feels on a particular topic, its correlation with more objective measures of positive and negative reactions remains a core focus of researchers in this domain (e.g., Armitage & Conner, 2000; DeMarree et al., 2014; Gebauer et al., 2013; Krosnick & Petty, 1995; Thompson et al., 1995). Methodologically, subjective ambivalence has been used as a tool to assess the performance of the many existing formulae designed to estimate an individual's ambivalence from his or her separate positive and negative reactions (Priester & Petty, 1996; Riketta, 2000; Thompson et al., 1995). Researchers have advanced a number of methods for calculating ambivalence in the past, but as noted none of these accounts for valence asymmetries. The common metric for evaluating one model versus another is to gauge the correlation between the estimate of objective ambivalence from a formula (a function of the perceiver's positive and negative reactions) and a person's subjective ambivalence, the logic being that a higher correlation implies a more valid and reliable assessment of objective ambivalence.

In general, previous work has established that the correlation between objective and subjective ambivalence tends to hover around .40 (e.g., Priester & Petty, 1996). The current studies uncovered correlations ranging from .14 to .60. However, the chief purpose of such a model or formula is to accurately estimate the experience it is meant to describe, and scholars have expressed surprise that this correlation is not higher. In many cases, the relatively low correlation has been explained by identifying other sources of subjective ambivalence, such as conflict between actual and desired attitudes (DeMarree et al., 2014), anticipated conflict (Priester et al., 2007), semantic incongruence (Gebauer et al., 2013), conflict between one's own and another's attitude (Priester

& Petty, 2001), and conflict between implicit and explicit evaluations (Petty, Tormala, Briñol, & Jarvis, 2006).

The current research suggests that negativity bias and positivity offset might also impact the strength of this link. In particular, if negative reactions carry more weight than otherwise equivalent positive reactions for mixed attitudes, and default positivity can emerge even when people have no positive information, these factors should be accounted for in any attempt to estimate a person's level of ambivalence based on his or her separate positive and negative reactions. At a minimum, it would seem important to adapt existing ambivalence formulae to incorporate weighting functions that capture the valence asymmetries delineated in the present work. If positive reactions carry less weight than negative reactions in shaping the experience of ambivalence for mixed attitudes, they should not carry the same weight in calculating ambivalence. Likewise, if positivity offset can shape the ambivalence people experience when they have univalent attitudes, ambivalence models should account for this in calculating an individual's objective ambivalence—for example, by adding a positive score even in the absence of positive reactions. We believe that modifying existing models or developing novel methods that capture these asymmetries could produce a reliable increase in the correlation between objective and subjective ambivalence, and would be of great interest and value to researchers in this domain.

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(Appendix follows)

Appendix

Traits From Studies 2 and 3 (Attitude Object: Frank)

Positive valence trait descriptors	Negative valence trait descriptors
Generous	Unreliable
Likable	Unpleasant
Cooperative	Irresponsible
Level-headed	Incompetent
Resourceful	Impolite
Patient	Irrational
Courteous	Shallow
Friendly	Irritable
Sensible	Egotistical
Tolerant	Lazy
Clever	Disagreeable
Enthusiastic	Cold

Antonym Traits (Supplemental Study in the General Discussion)

Positive valence trait descriptors	Negative valence trait descriptors
Thoughtful	Unfocused
Eloquent	Inarticulate
Likeable	Deceptive
Sincere	Insincere
Dependable	Selfish
Caring	Boring
Loyal	Power-Hungry
Effective	Irresponsible
Diligent	Hypocritical
Genuine	Dishonest
Reliable	Negligent
Trustworthy	Unpredictable

Tablet Descriptors (Supplemental Study in the General Discussion)

Positive valence attribute descriptors	Negative valence attribute descriptors
Vivid display	Lackluster screen
Lightweight	Heavy
Durable	Flimsy
Stylish	Ugly design
Intuitive touchscreen	Finicky touchscreen
Good speakers	Bad speakers
Great wireless connectivity	Crappy wireless
Video is high quality	Low quality video
Long battery life	Surprisingly short battery life
Simple to use connector	Difficult charger
Impressive warranty	Unimpressive guarantee
Lots of storage space	Not enough storage space

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